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Paleomagnetism of the Rio Perdido mafic dike swarm (1110 Ma) and the paleogeography of Amazonia and its role for supercontinents Rodinia and Gondwana

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ABSTRACT

The Rio Apa Terrane (RAT) is situated at the south end of the Amazonian Craton (AC) separated from it by the aulacogen Tucavaca belt. It encompasses basement rocks that evolved mainly during the Paleoproterozoic. Its relationship to the AC during Proterozoic is yet undefined, although geologic, geochronologic and isotopic evidence suggest it was close to the Ventuari-Tapajós and Rio Negro-Juruena Provinces at Paleoproterozoic and linked to southwestern AC at ca. 1320 Ma. Aiming to elucidate the paleogeographic relation of the RAT to the AC during the Proterozoic, we performed a paleomagnetic study of the 1110 Ma Rio Perdido mafic dike swarm that crosscut the RAT. These dikes strike preferentially WNW-ESE and are mostly undeformed and unmetamorphosed. Magnetic experiments (hysteresis curves, isothermal remanent magnetization (IRM) curves, thermomagnetic curves and first order reverse curves-FORCs) revealed pseudo-single domain (PSD) Ti-poor titanomagnetite as the main magnetic carrier in these dikes. Alternating field (AF) and thermal demagnetization were efficient in isolating southwestern and northeastern directions with low inclinations, whose mean direction (Dec = 51.8°, Inc = −0.8°, N = 14, α_{95} = 13.8°, k = 9.3) yielded the paleomagnetic pole (RP) at 34.8°S/197.3° (A95 = 10.9°). This pole passed a baked contact test and meets 6 out of 7 pole reliability criteria (R = 6). The Rio Perdido pole implies that RAT could be linked to the southwestern AC at 1110 Ma but rotated counterclockwise ~ 110° related to its present position. The AC linked to RAT probably belonged to the 1100 Ma Umkondia megacontinent which later collided with Laurentia during the formation of Rodinia at ca. 1000–900 Ma. Also, paleomagnetic data suggest that the RAT and the southwestern AC (Paraguá Terrane) were linked to Ventuari-Tapajós and Rio Negro-Juruena provinces at ca. 1780 Ma and had a joint geological evolution during the Paleo- and Mesoproterozoic. Finally, during the amalgamation of West Gondwana, the RAT rotated clockwise to its present position.

1. Introduction

The Rio Apa Terrane (RAT) is a ca. 220 km exposure along the N-S direction and 60 km wide, mainly composed of Paleoprotezoic rocks (Teixeira et al., 2020). It is located at the south end of the Amazonian Craton (Fig. 1). Its tectonic relationship with the Amazonian Craton has been intensively debated, as it is truncated by the Brasiliano Tucavaca Belt in Gondwana times. From the tectonic-geochronological divisions presented by Cordani et al. (1979), Litherland and Bloomfield (1981) and Litherland et al., (1986,1989), the RAT was excluded from the Amazonia reconstructions, assuming the Tucavaca Belt as a marginal belt of aulacogen nature. Alternatively, Ruiz et al. (2005) considered that the Amazonian Craton extends to the Rio Apa region based on geological evidence, such as: (i) continuity of the Paraguay Belt to the south, around the RAT; (ii) the stratigraphic correlation of the Tucavaca sedimentary rocks and those exposed in the Paraguay Belt, and (iii) the

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Tucavaca Belt most probably represents an aborted branch of a triple junction centered at the Rincón del Tigre/Corumbá region. So, the inclusion of this terrane in the Amazonian Craton would require a new approach in the formulation of Precambrian paleogeographic reconstruction models.

Similarly, recent advances in geological and geochronological studies led several authors to consider the RAT as a fragment of the Amazonian Craton. Most authors subdivide the RAT into two tectonic domains (Fig. 2): the (2100–1820 Ma) Western Domain and the (1800–1720 Ma) Eastern Domain (e.g., Cordani et al., 2010; Lacerda-Filho et al., 2016; Faleiros et al., 2016; Ribeiro et al., 2020; 2022a; Teixeira et al., 2020). According to Ribeiro et al. (2020), zircon U–Pb–Hf

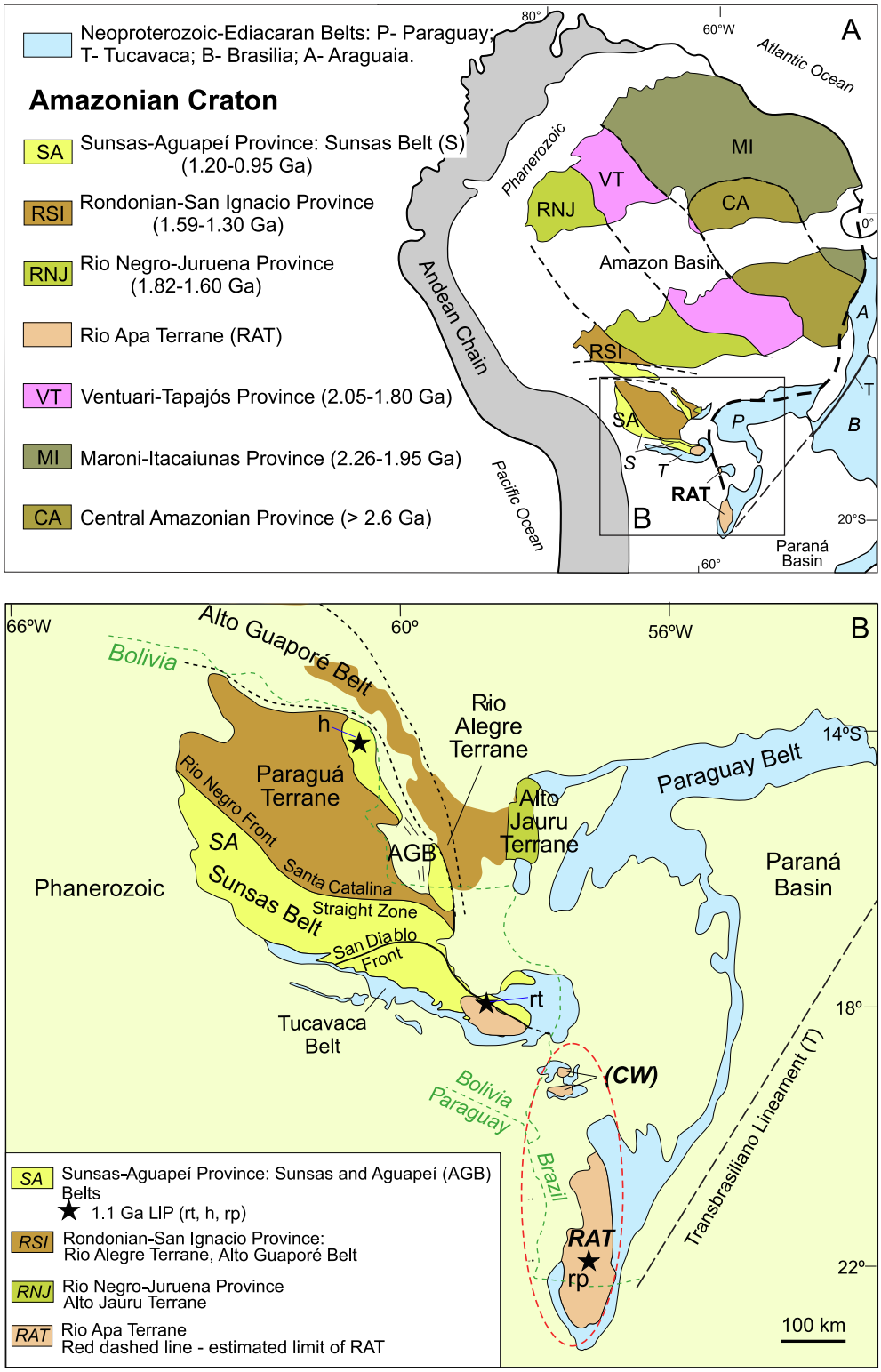


Fig. 1. (A) Amazonian Craton and their tectonic provinces, with the inclusion of the Rio Apa Terrane (RAT). (B) SW Amazonian Craton, showing the RAT and correlative rocks (CW – Corumbá window). AGB (Aguapeí fold and thrust belt). Stars in the SW Amazonian Craton represent 1110 Ma LIP components: h (Huanchaca sills and dikes; rt (Rincón del Tigre complex); rp (Rio Perdido dikes). Adapted from Teixeira et al. (2020) and Ribeiro et al., (2022a).

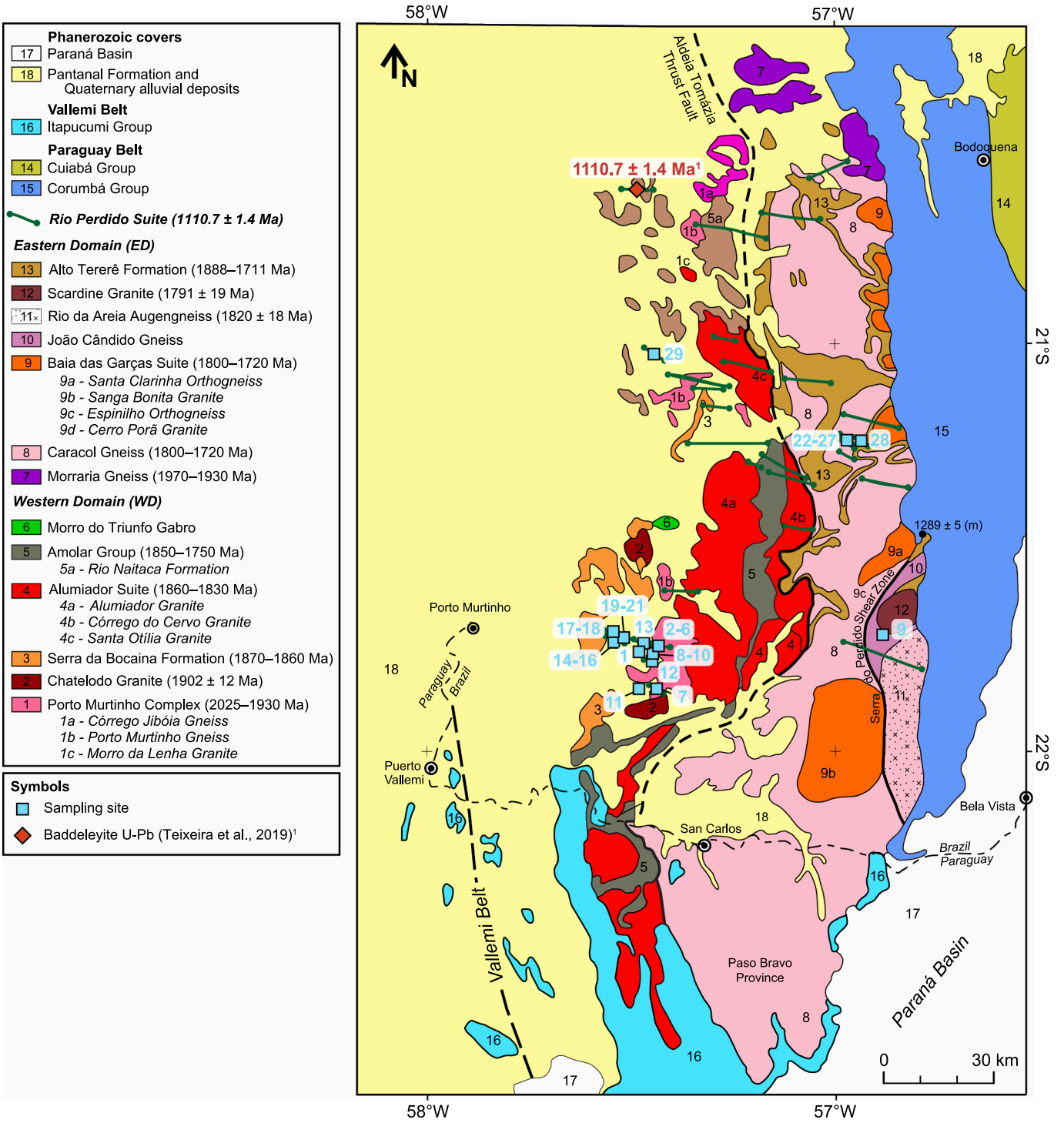


Fig. 2. Geologic map of the Rio Apa Terrane. Adapted from Ribeiro et al., (2022a,b). Paleomagnetic sampled sites (full black squares) are numbered according to Table 1. The U-Pb (Baddeleyiite) dated dike is also shown as a lozenge.

isotopic data suggest that the RAT is a result of long-lived evolution which is marked by a spatial transition from 2100 to 1840 Ma crustal reworking in the Western Domain to more juvenile magmatism spanning 1800–1720 Ma in the Eastern Domain. This transition is interpreted to reflect convergent margin magmatism associated with periods of subduction zone advance and retreat in an accretionary orogenic setting. Such a polycyclic history compares well with those of Ventuari-Tapajós and Rio Negro-Juruena Provinces in the Amazonian Craton (Ribeiro et al., 2020), as well with that of the Paraguá Terrane in the counterpart of Bolivia. Consequently, the primary link between the RAT and the Amazonian Craton is reinforced (e.g., Cordani et al., 2010; Faleiros et al., 2016; Ribeiro et al., 2020; 2022a; Teixeira et al., 2020; Redes et al., 2020). From a paleotectonic perspective, the RAT was a close neighbor to Amazonia during Paleo- to Mesoproterozoic times. Noteworthy, the RAT is crosscut by the well-dated 1110 Ma Rio Perdido dike swarm (Teixeira et al., 2019) while Paraguá Terrane (Amazonian Craton) is crosscut by the well-dated 1110 Ma Rincón del Tigre mafic–ultramafic complex and by the Huanchaca Intrusive Suite that includes mafic sills and dikes of identical age (Teixeira et al., 2015). The large geographic extent of such magmatic events is consistent with the

existence of a large igneous province (LIP) at that time. During the Gondwana formation, the RAT amalgamated to the Amazonian craton along with the Tucavaca Belt aulacogen.

This paper presents paleomagnetic results obtained for the Rio Perdido dike swarm with important implications for the relative position of the RAT to the Amazonian Craton at 1110 Ma. Furthermore, we discuss the correlation of the Western and Eastern domains (RAT) with the Ventuari-Tapajós and Rio Negro-Juruena provinces given the available geological, isotopic and paleomagnetic data, their paleogeography at 1100 Ma relative to other cratonic-forming blocks of Supercontinent Rodinia, and its participation in the formation of Gondwana.

2. Geological setting

The RAT is composed of Paleoproterozoic medium-grade, *meta*-igneous rocks and low- to medium-grade metasedimentary sequences, intruded by granitic rocks (Cordani et al., 2010, Lacerda-Filho et al., 2016; Faleiros et al., 2016; Teixeira et al., 2020; Ribeiro et al., 2022a, 2022b), that extends from the central-west Brazil to the northeast Paraguay (Figs. 1, 2). It is covered by the Neoproterozoic-Cambrian pelitic-carbonatic rocks from the Ediacaran Corumbá Group, and on the west side, by Quaternary rocks (Teixeira et al., 2020). A small basement window of the RAT is exposed to the NW of the RAT, in between the Corumbá rocks (Fig. 1B). A major thrust system known as Aldeia Tomázia Thrust Fault (after Ribeiro et al., 2022a) occurs between the Western and the Eastern domains (Fig. 2), which is characterized by a tectonic transport moving to NWW (Araújo et al., 1982; Faleiros et al., 2014; 2016; Lacerda Filho et al., 2006; 2016; Remédio and Faleiros, 2014; Cordani et al., 2010; Teixeira et al., 2020; Ribeiro et al., 2020, 2022a, b).

The Western Domain is formed by a gneissic basement intruded by voluminous calc-alkaline granitoids and structurally overlain by a lower-grade metasedimentary succession (e.g., Cordani et al., 2010; Faleiros et al., 2016; Lacerda-Filho et al., 2016; Teixeira et al., 2020; Ribeiro et al., 2020; 2022a). The Porto Murtinho Complex represents the oldest (2025–1930 Ma) unit, composed of mylonitic, grey, monzonitic orthogneiss and undeformed, green, porphyritic monzogranite. Granitoids of the Alumiador Suite and the felsic volcanic units (medium to high-K, calc-alkaline rhyolite) from the Serra da Bocaina Formation compose the 1870–1820 Ma Amoguijá Belt (Fig. 2) (Lacerda-Filho et al., 2016), interpreted as a continental magmatic arc derived primarily from crustal-recycled rocks (Manzano et al., 2012; Godoy et al., 2014; Lacerda-Filho et al., 2016; Teixeira et al., 2020). These basement rocks are covered by a succession of volcano-sedimentary rocks metamorphosed in the greenschist facies, the Amolar Group (Cordani et al., 2010) and the Rio Naitaca Formation (Faleiros et al., 2016). A *meta*-andesite at the base of the Rio Naitaca Formation was dated (U–Pb) at 1813 ± 18 Ma (Faleiros et al., 2016), and a *meta*-basalt hosted by metasandstone from the same formation yielded a zircon U–Pb upper intercept age of 1819 ± 10 Ma (Teixeira et al., 2020). Coherent detrital zircon U–Pb–Hf isotopic data from the Amolar Group and Rio Naitaca Formation demonstrate an autochthonous origin in relation to the Western Domain (Ribeiro et al., 2020), precluding a direct link with the supracrustal rocks from the Eastern Domain.

The Eastern Domain includes the 1770–1720 Ma Caracol gneissic rocks and the 1750–1720 Ma Baía das Garças granites that are tectonically ascribed as the Caracol magmatic arc, dominantly characterized by juvenile rocks (Teixeira et al., 2020; Ribeiro et al., 2020). This magmatic arc includes the 1770 Ma metavolcanic-sedimentary rocks from the Alto Tererê Group, which evolved in a back-arc environment (Lacerda-Filho et al., 2016; Ribeiro et al., 2020). The Paso Bravo unit represents part of the Caracol Arc, while the Morraria orthogneiss is currently interpreted to represent a potential fragment of the Western Domain entrapped in the Eastern Domain, likely during the retreat episode of the accretionary orogen (Ribeiro et al., 2020).

According to recent geological models, the RAT resulted from a long-lived, west-dipping subduction-related evolution, experiencing advancing and retreating episodes from ca. 2100 Ma to 1720 Ma (e.g., Ribeiro et al., 2020; 2022a; 2022b). A similar geotectonic model has been attributed to the Southwestern Amazonian Craton, supported by geophysical evidence (Trevisan et al., 2021; Motta et al., 2022; Ribeiro et al., 2022a). This fact suggests a joint geological evolution of RAT and Paraguá Terrane during the Paleoproterozoic (e.g., Ribeiro et al., 2022a).

The metamorphic evolution during Mesoproterozoic times is an important link for the final reconstruction of the growing SW Amazonian margin. Ribeiro et al., (2022b) presented a robust petrochronological study demonstrating that the tectonothermal events present in the Paraguá Terrane are also found in the RAT. This demonstrates that they belong to a paired metamorphic belt with Paraguá Terrane representing the high-temperature domain (closer to the accretionary margin of Amazonia). The Rio Apa Terrane, in turn, represents the high-pressure/medium-temperature domain (further from the margin). When combined with the zircon Hf-time series, Ribeiro et al. (2022a; 2022b) demonstrated that the Rio Apa and Paraguá terranes belong to the same accretionary margin since Paleoproterozoic times. To summarize, there is very robust and compelling evidence of a petrogenetic link between Paraguá, Rio Apa and SW Amazonian Craton.

2.1. Rio Perdido dike swarm

The Rio Perdido Suite mafic dike swarm, the focus of our study, named previously as the Rio Perdido Intrusive Suite (RPS) by Lima et al. (2017), intrude all geological units of the RAT in Brazil and Paraguay (Fig. 2). This intrusive suite comprises massive, greenish gray to dark gray dikes, varying from very fine- to fine-grained diabase and fine- to medium-grained microgabbro. The Rio Perdido dikes are sub-vertical to vertical (80–90°), trending to N70°-90°E and N70°-90°W (Lima, 2016). They are between 1 and 30 m thick and 100–200 km long, with abrupt and discordant contact with the regional host rock trends and without deformation and metamorphism (Lima, 2016; Faleiros et al., 2016; Teixeira et al., 2019). Precise baddeleyite U-Pb (ID-TIMS) dating on four fractions of a dike from this swarm (Fig. 2) yielded an age of 1110.1 ± 2.3 Ma, interpreted as the crystallization age (Teixeira et al., 2019). This dike is situated in an indigenous reservation (Kadiwéu) of difficult access, and could not be sampled for our paleomagnetic study (see below).

3. Sampling and procedures

A total of 219 oriented drill cores (from 26 sites) and 12 oriented hand samples (from three sites) were sampled from 29 mafic dikes from the Rio Perdido swarm (Table 1), which are located to the east and northeast of the Porto Murtinho City (Fig. 2). Site 22 is represented by a well-exposed, 2 m thick dike (Table 1), which cuts the Caracol banded gneissic rocks. Eight oriented cores were also sampled from the gneissic rocks at the contact and up to 6 m from the dike's contact for a baked contact test. Cylindrical cores were drilled using a portable, gasoline-powered rock drill. The samples were oriented by magnetic and sun compasses whenever possible.

All magnetic measurements were performed at the Laboratory of Rock Magnetism and Paleomagnetism (*USPmag*), University of São Paulo, Brazil. Cylindrical specimens of 2.2 × 2.5 cm were cut from each sample using a non-magnetic saw and adequately marked for laboratory treatments. Conventional stepwise thermal and alternating field (AF) demagnetization were used to separate magnetic components carried out by the samples. A generical AF demagnetization routine with steps of 2.5 mT (up to 15 mT), 5 mT (15–50 mT), and 10 mT (50–100 mT) was employed for AF demagnetization using a tumbler AGICO AF demagnetizer or an automated three-axis AF demagnetizer coupled to a 2G-Enterprises cryogenic magnetometer. Steps of 50 °C (100–500 °C) and 20 °C (500–700 °C) were used for thermal demagnetization in a Magnetic Measurements TD-48 oven.

Table 1
Paleomagnetic data for the Rio Perdido dike swarm.

Sites	GLat (S)/GLong (W)	Strike	Width (m)	Samples	N/n	Site mean direction				VGP	
						Dec (°)	*Inc* (°)	α_{95} (°)	*K*	*Plat* (°N)	*Plong* (°E)
1	21°43′12.1″/ 57°26′42.1″	N85W	———	SR1A-1 J	16/13	40.6	−13.3	6.1	46.7	−48.1	197.8
2	21°42′20.4″/ 57°25′28.4″	E-W	———	SR2A-2 J	———	———	———	———	———	———	———
3#	21°42′20.7″/ 57°25′28.5″	E-W	2.5	SR3A-3 J	10/10	4.8	−21.7	6.7	52.6	−78.6	147.1
4	21°42′22.5″/ 57°25′29.1″	———	———	SR4A-4E	———	———	———	———	———	———	———
5#	21°42′22.5″/ 57°25′29.1″	N80W	25	SR5A-5E	10/10	359.5	−24.6	7.6	41.3	−81.2	119.4
6#	21°42′16.2″/ 57°25′25.5″	N85W	———	SR6A-6G	12/12	5.2	−37.2	4.4	97.9	−85.1	203.5
7	21°49′59.9″/ 57°26′02.3″	N80W	30	SR7A-7L	———	———	———	———	———	———	———
8	21°44′35.1″/ 57°26′46.2″	E-W	30	SR8A-8I	15/11	54.5	−7.8	7.3	40.6	−34.3	202.0
9	21°39′16.2″/ 56°52′54.2″	N85W	———	SR9A-9H	10/10	62.3	1.4	5.0	94.8	−25.3	201.4
10	21°44′37.6″/ 57°26′48.6″	———	———	SR10A-10F	10/10	41.4	−16.7	9.8	25.5	−48.1	201.7
11	21°44′42.4″/ 57°25′47.7″	———	———	SR21A-21K	15/09	16.6	−8.9	8.3	39.4	−64.4	167.9
12	21°44′24.8″/ 57°25′48.6″	———	———	SR22A-22I	10/06	16.7	−12.0	11.2	37.0	−67.5	170.8
13	21°43′04.2″/ 57°26′39.3″	N70W	20	SR23A-23I	———	———	———	———	———	———	———
14	21°41′40.1″/ 57°31′18.3″	———	———	SR24A-24G	11/08	71.6	−20.6	6.4	76.3	−20.9	216.0
15	21°41′40.2″/ 57°31′12.0″	N80W	8	SR25A-25G	10/10	70.7	–33.4	11.0	20.2	−24.0	223.5
16	21°41′44.6″/ 57°31′15.5″	N85W	4	SR26A-26L	———	———	———	———	———	———	———
17	21°41′10.7″/ 57°31′23.7″	N86E	5	SR27A-27I	11/09	256.2	−2.0	10.0	27.5	−12.8	207.2
18	21°41′07.0″/ 57°31′24.2″	N87E	———	SR28A-28H	———	———	———	———	———	———	———
19	21°41′42.9″/ 57°30′10.9″	N87W	6	SR29A-29H	———	———	———	———	———	———	———
20	21°41′33.0″/ 57°30′09.4″	———	———	SR30A-30I	———	———	———	———	———	———	———
21	21°41′19.0″/ 57°31′34.7″	N82W	35	SR31A-31H	———	———	———	———	———	———	———
22	21°10′02.7″/ 56°57′48.6″	E-W	2	SR32A-32H	07/06	219.7	−19.4	8.4	64.0	−40.1	178.4
22^{+}	21°10′02.7″/ 56°57′48.6″			SR32I-32P	17/10	199.4	4.4	11.0	19.4	−63.4	170.0
22^{++}	21°10′02.7″/ 56°57′48.6″			SR32I-32P	17/07	305.4	−39.5	8.1	55.9	−39.6	20.9
23	21°10′01.2″/ 56°57′47.1″	N80W	7	SR33-36	12/10	235.7	−13.9	15.8	10.3	−28.5	191.9
24	21°10′01.9″/ 56°58′52.1″	N76W	1.5	SR37-40	16/12	241.8	−31.0	3.0	192.3	−18.5	185.9
25	21°10′01.6″/ 56°57′55.1″	N82W	———	SR41-44	10/4	255.0	−26.6	5.2	314.1	−8.4	194.3
26	21°10′02.2″/ 56°58′01.4″	N75W	———	SR45A-45F	———	———	———	———	———	———	———
27	21°10′03.7″/ 56°57′58.5″	N70W	———	SR46A-46F	———	———	———	———	———	———	———
28#	21°10′43.9″/ 56°55′36.8″	N70W	50	SR47A-47H	12/9	356.3	−28.8	7.2	51.8	−83.2	91.3
29	20°56′48.3″/ 57°24′57.3″	———	———	SR48A-48L	12/11	47.3	8.2	13.3	12.8	−37.3	189.7
				Mean	18	41.5	−7.8	15.2	6.1		
				Pole				13.2	7.9	−46.2	195.2
				Mean*	14	51.8	−0.8	13.8	9.3		
				Pole*	RP			10.9	14.3	−34.8	197.3

*N/n – number of specimens measured/used in the mean; Dec - declination, Inc - inclination, α_{95} e K – Fisher's statistical parameters, Plat - Paleolatitude, Plong - Paleolongitude e VGP – virtual geomagnetic pole. * Site mean direction after using a 40° cutoff (# - Eliminated sites), and the respective paleomagnetic pole.* ***22***– *Site sampled for the baked contact test;* ***22***$^{+}$*- basement rocks baked by the dike;* ***22***$^{++}$*- unbaked basement rocks; Thickness and strike were measured whenever visible in the field.*

Measurements of remanent magnetization were carried out using a spinner JR6 magnetometer (AGICO) or a 2G-Enterprises cryogenic magnetometer. Both apparatuses are housed in a shielded room with an internal field smaller than 500 nT. Magnetic components were identified and calculated using orthogonal projections (Zijderveld, 1967) and principal component analysis (Kirshvink, 1980), respectively. At least four demagnetization steps were used to calculate vectors from least-squares fits, and only samples with mean angular deviation (MAD) smaller than 11° were retained. For some components where magnetic intensity does not decrease significantly during AF or thermal demagnetization, the Characteristic remanent magnetization (ChRM) was calculated including the origin, and/or anchored to the last point, where it is supposed that ChRM is the only direction remaining in the rock. Fisher's (1953) statistics were used to calculate site mean directions and paleomagnetic poles.

We studied the magnetic mineralogy of selected samples using different methods. Magnetic hysteresis and isothermal remanent magnetization (IRM) curves were measured with a MicroMag-VSM device (Model 3900, Princeton Measurements Corporation). We also used the same device to obtain First Order Reversal Curves (FORCs, Roberts et al., 2000). A Python-based machine learning tool (*FORCsensei*, Heslop et al., 2020) was applied to calculate the FORC distributions. We pre-processed the data by removing artifacts from the first and last points, and by correcting for (i) drift and (ii) slope (at 70 %). Smoothing regions (in terms of size and shape) were adjusted depending on their position in the FORC diagram (*VARIFORC* - Egli, 2013) using a local weighting scheme: (i) the minimum horizontal and vertical smoothing values ($S_{c,0}$ and $S_{u,0}$, respectively), (ii) the minimum horizontal smoothing away from $B_c = 0$ ($S_{c,1}$), (iii) the minimum vertical smoothing away from a possible central ridge ($S_{u,1}$) and the (iv) linear increase in smoothing with increasing field (λ). We use the parameter Ψ to check how well the local data matches second-order polynomials at each point for a specific set of *VARIFORC* smoothing parameters (it will be 1 in this case) or to detect underfitting or overfitting in some areas (Heslop et al., 2020). Lastly, thermomagnetic curves were performed with a CS3 furnace coupled to a Kappabridge KLY-4S instrument (AGICO, Inc, Brno). Measurements of temperature (°C) per susceptibility (κ) (in both low and high temperatures) were performed, also using a constant argon flow during the heating cycle to avoid excessive oxidation.

Finally, we accessed the magnetic fabric of 355 specimens (from sites 11 to 25 – Table 1) through an Agico Kappabridge MFK1-FA, individually acquiring their main susceptibility directions ($\kappa_1 \geq \kappa_2 \geq \kappa_3$). Jelínek's AMS parameters (Jelínek, 1977, 1981; Jelínek, 1978) were used to investigate the anisotropy degree $\left(P = \kappa_1 \bullet \kappa_3^{-1}\right)$ and the shape $\left(T = [2 \bullet ln(\kappa_2) - ln(\kappa_1) - ln(\kappa_3)] \bullet [ln(\kappa_1) - ln(\kappa_3)]^{-1}\right)$ of each ellipsoid formed by the main susceptibilities.

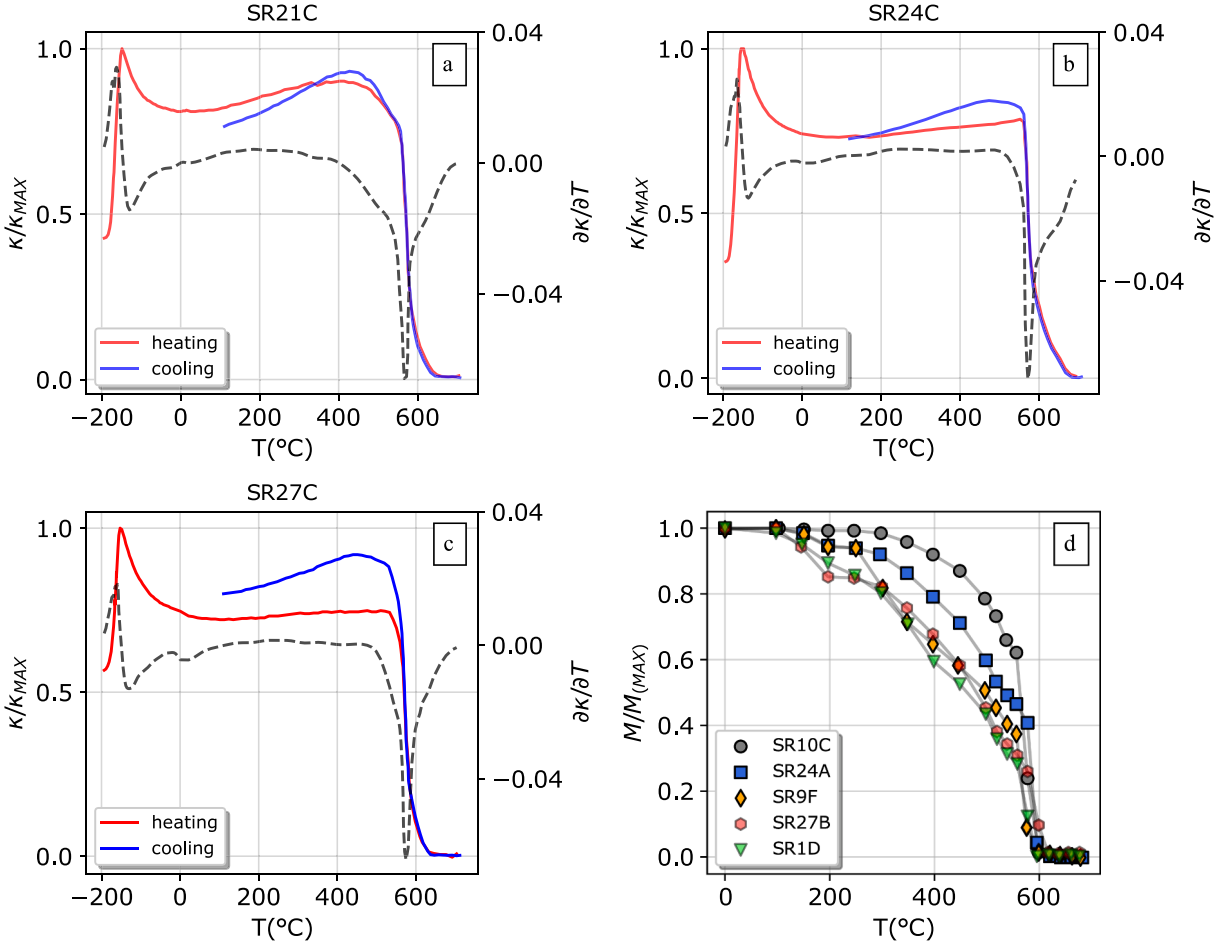


Fig. 3. Thermomagnetic curves (T × k) of representative samples from the dikes (a-c). The red curve is the heating curve, while the blue one is the sequential cooling. The dashed line is the gradient of the susceptibility, attesting to two major features: the low-temperature Verwey transition and the Curie temperature of magnetite. d) Examples of magnetization decay through a thermal demagnetization procedure are shown. These examples indicate that Ti-poor magnetite is the main ferrimagnetic mineral in these rocks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Results

4.1. Magnetic mineralogy

All the thermomagnetic curves demonstrate similar features in the heating curves (Fig. 3a to c), which are well demarked by the gradient curves: (i) an increase in magnetic susceptibility near the Verwey transition (-155 °C) and its drop around the Curie temperature (≈ 580 °C) indicating the presence of Ti-poor titanomagnetite. The evidence for such mineral is also found on typical normalized intensity curves for samples that yielded stable directions after thermal treatment (Fig. 3d). Usually, these samples bear unblocking temperatures (Tub) distributed up to ca. 550 °C, when a steep drop in intensity occurs towards 600 °C.

Hysteresis, IRM and FORC curves were performed for selected samples whose results may be viewed in Supplementary-File 1. Magnetic hysteresis of all the samples shows narrow-waist loops with a small paramagnetic contribution (Fig. 4 a, b). Most of the artificial remanence in IRM curves is acquired before 100 mT (Fig. 4c, gradient curves), while saturation is reached at fields between 100 and 300 mT (Fig. 4c). Both experimental techniques indicate that magnetite is the main remanence carrier in these rocks.

Mrs/Ms (saturation remanent magnetization/saturation magnetization) and Bcr/Bc (coercivity of remanence/bulk coercive force) ratios from magnetic hysteresis and IRM curves fall within the pseudo-single-domain (PSD) field in the Dayplot (Day et al., 1977) diagram (Fig. 4d). Most of the samples are well adjusted in a trend following linear mixtures of SD + MD populations of grains (Dunlop, 2002), the coarse grains most likely being responsible for the low coercivity unstable components observed for some samples in our AF demagnetization procedure (as it will be further discussed). Assessing the magnetic domain state of magnetic carriers using the Day plot is a long-established methodology in paleomagnetic studies, but it has also been suggested that it may offer misguided insights (Lascu et al., 2018). FORC diagrams might offer a more robust approach in this matter.

As expected from the hysteresis curves, FORCs diagrams (Fig. 4*e, f*) are similar for the two measured specimens (SR9B (site 9) and SR25C (site 15) – the position of these samples in the Day's diagram are shown in Fig. 4d). Optimization of smoothing parameters using *FORCsensei* resulted in well-fit data to second order polynomials $\Psi \geq 0.69$, which represents only a few non-representative under fitted regions, and small smoothing factors ($S_{c,0}, S_{c,1}, S_{u,0}, S_{u,1} = 2$). The diagrams show that most of the magnetic contribution is located at low coercivity values at the $B_c((B-B_r)/2)$ axis, but far from the origin. This is consistent with the PSD behavior (vortex state) described by Roberts et al. (2000), which combines features of both MD-like and SD-like magnetic moments. The typical PSD FORC diagram of fine-grained synthetic magnetite has a low coercivity center and a FORC distribution that spreads towards the B_u axis ($(B+B_r)/2$) (Muxworthy and Dunlop, 2002). When the grain size in the PSD range increases, the SD-like peaks shift to the lowest coercivities and the outer contours diverge more towards $B_u = 0$ (more MD-like) (Roberts et al., 2000). PSD grains have been numerically demonstrated to be highly stable magnetic recorders of Earth's magnetic field (Nagy et al., 2017; Lascu et al., 2018), which supports the paleomagnetic data presented in this paper.

AMS data are presented in Supplementary-File 2 (Excel file) comprising two sheets. The first sheet contains all the data obtained for each individual specimen, while the second sheet displays the compiled Jelinek (1977; 1978) statistical values for the sample groups we were able to analyze. From these data, it is observed that bulk magnetic susceptibility ($\kappa_{average}$) varies from $1.5 \bullet 10^{-3}$ to $5.2 \bullet 10^{-2}$, with an average value of $2.2 \bullet 10^{-2} \pm 1.29 \bullet 10^{-2}$. Although there is a very well-marked magnetic fabric, the mean anisotropy degree is low ($P = 1.05 \pm 0.024$), not showing specific trends when plotted against bulk magnetic susceptibility ($\kappa_{average}$) (Fig. 5a). Ellipsoids are almost equally distributed along prolate and oblate fields in the Jelínek plot (Fig. 5b), with no obvious correlation between P and the shape parameter (T). Although there is some spreading, κ_1 density contours (Schmidt method, 1 % area) indicate intermediate to high plunges (Fig. 5c). κ_3 directions are more spread (which could be related to magma flow), but its greater density is in the NW sector (Fig. 5d). Since there is no mineralogical evidence for an inverse fabric as the mineralogical data indicates mostly a PSD state, we can interpret κ_3 as a pole of the magnetic foliation of the dikes and, therefore, as an indicative of strikes along the ENE-WSW quadrant (which agrees with the main regional trends for the Rio Perdido dikes as explained in section 2). Previous petrographic analysis indicates no evidence of metamorphism and deformation of these rocks (Lima, 2016; Lima et al., 2017). When this information is paired with the magnetic mineralogy experiments, we imply that magnetite is probably of primary origin.

4.2. Paleomagnetism

AF treatment efficiently demagnetized the studied samples. After the elimination of random secondary components with coercivities up to 10 to 30 mT (Fig. 6; Table 1) a NE (and an antipodal SW) component with low inclination was obtained for samples from 18 of the 28 analyzed dikes.

Thermal demagnetization was applied to two or three samples from each site to disclose unblocking temperatures associated with the magnetic minerals. Normally, it also separated similar NE (and the antipodal SW) directions, after secondary components were eliminated at temperatures up to 500–520 °C (Fig. 6). Samples from the other 10 dikes yielded either unstable behavior or inconsistent directions and were henceforth discarded. Secondary and low coercivity components commented here could be yielded by less stable MD particles within the rock fabric while the most stable ones would be carried by PSD particles, as is also suggested by the observed trend of Mrs/Ms and Bcr/Bc ratios on the Day plot diagram (Fig. 4d).

Paleomagnetic calculations were further performed using the *PmagPy* package (Tauxe et al., 2016). The mean directions of 18 sites (after flipping the polarity and declination of SW components) resulted in a cluster at Dec = 41.5°, Inc = -7.8° ($\alpha_{95} = 15.2°$, k = 6.1, Fig. 7b), and their mean VPGs positions resulted in a paleomagnetic pole at 46.2°S/195.2°E (Table 1, A95 = 13.2°, K = 7.9).

To account for statistical significance, Vandamme's cutoff (Vandamme, 1994) was applied, but no VPGs were excluded from such procedure (Fig. 7c). Instead, applying a 40° cutoff resulted in the exclusion of four VGPs (Fig. 7d), whose main site directions were nearly North centered. An additional justification to exclude these sites from the mean is that their mean directions are close to the present geomagnetic field. Using 14 of the VGPs within the 40° cutoff resulted in a new cluster at Dec = 51.8°, Inc = -0.8° ($\alpha_{95} = 13.8°$, k = 9.3, Fig. 7e). The new paleomagnetic pole (RP, Fig. 7f) is located at 34.8 °S / 197.3 °E (Table 1, A95 = 10.9°, K = 14.3), which passes the Deenen's criteria for paleosecular variation (Deenen et al., 2011, 2014), since $A95_{min}$ (4.2°) < $A95_{RP}$ (10.8°) < $A95_{max}$ (15.6°).

We conducted two reversal tests to assess whether the site mean directions with opposite polarities observed in the dikes (NE and SW) are antipodal: the classic McFadden and McElhinny (1990) reversal test and the Common True Mean Directions (CTMD) test available at *paleomagnetism.org* (Koymans et al., 2016; Tauxe, 2010). For that, we flipped the SW directions to the NE quadrant and inverted their inclination. The results of both tests were negative. In the McFadden and McElhinny (1990) test, the observed angle (33.76°) exceeded the critical angle ($y_c = 23.34°$). In the CTMD test, although there was overlapping for the X and Y components, no overlapping was observed for the Z component (Fig. 8). Both tests suggest that despite the α_{95} of the mean directions for these two groups overlapping (Fig. 8d), the directions are too dispersed to be considered the same with 95 % confidence.

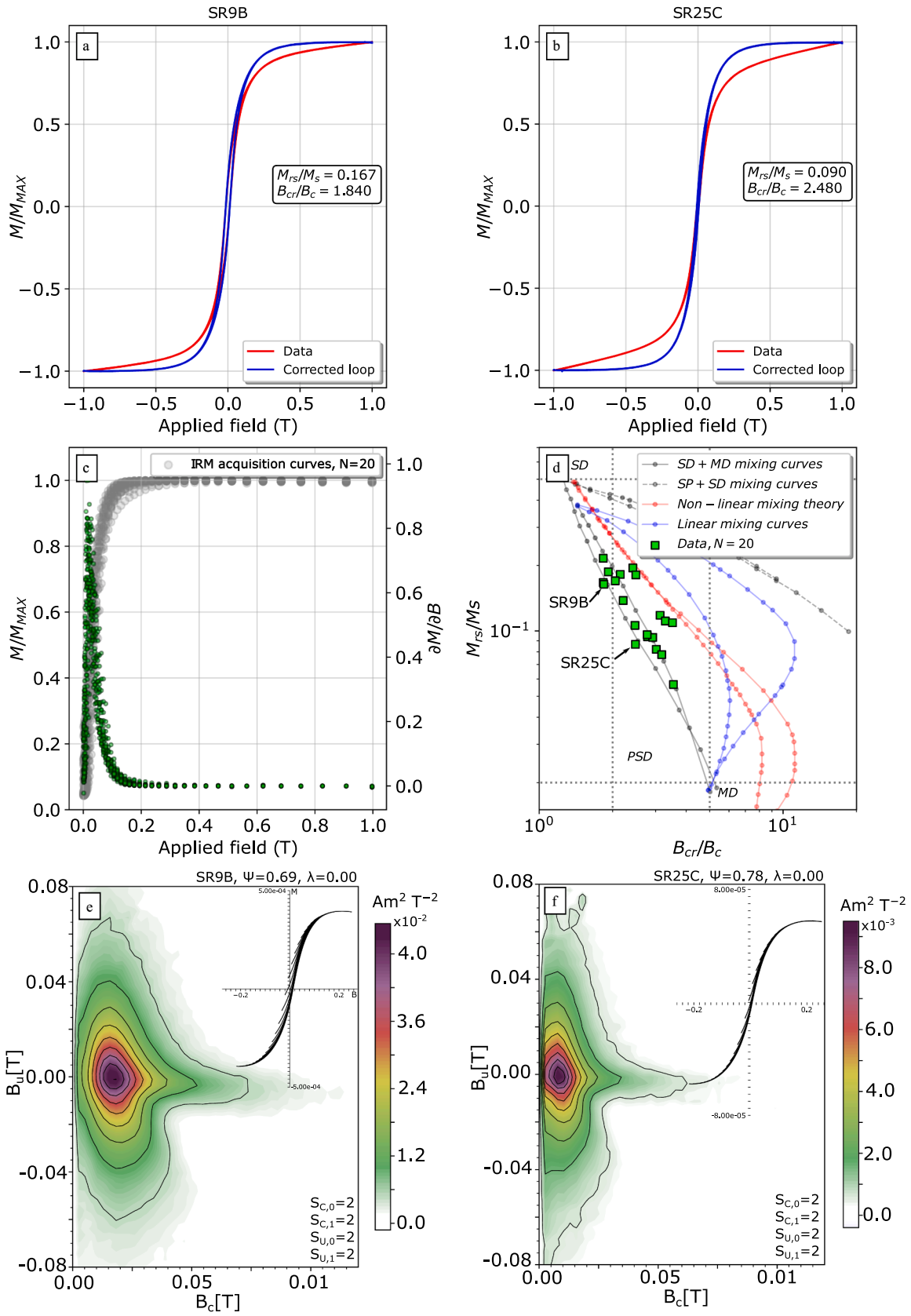


Fig. 4. Examples of magnetic hysteresis (a, b) showing narrow-waist loops. The red curves and blue curves are the uncorrected and paramagnetic-corrected loops (respectively). c) IRM acquisition curves and their gradient (green dots), showing that most of the remanence is acquired before 100 mT while saturation is sequentially reached in low fields. d) Mrs/Ms × Bcr/Bc data on the Day plot (Day et al., 1977) together with theoretical and experimental mixture curves (Dunlop, 2002). e, f) FORC diagrams for samples SR9B and SR25D (displayed in Fig. 4d) show typical contributions of populations of PSD grains (parameters are explained in Section 3). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

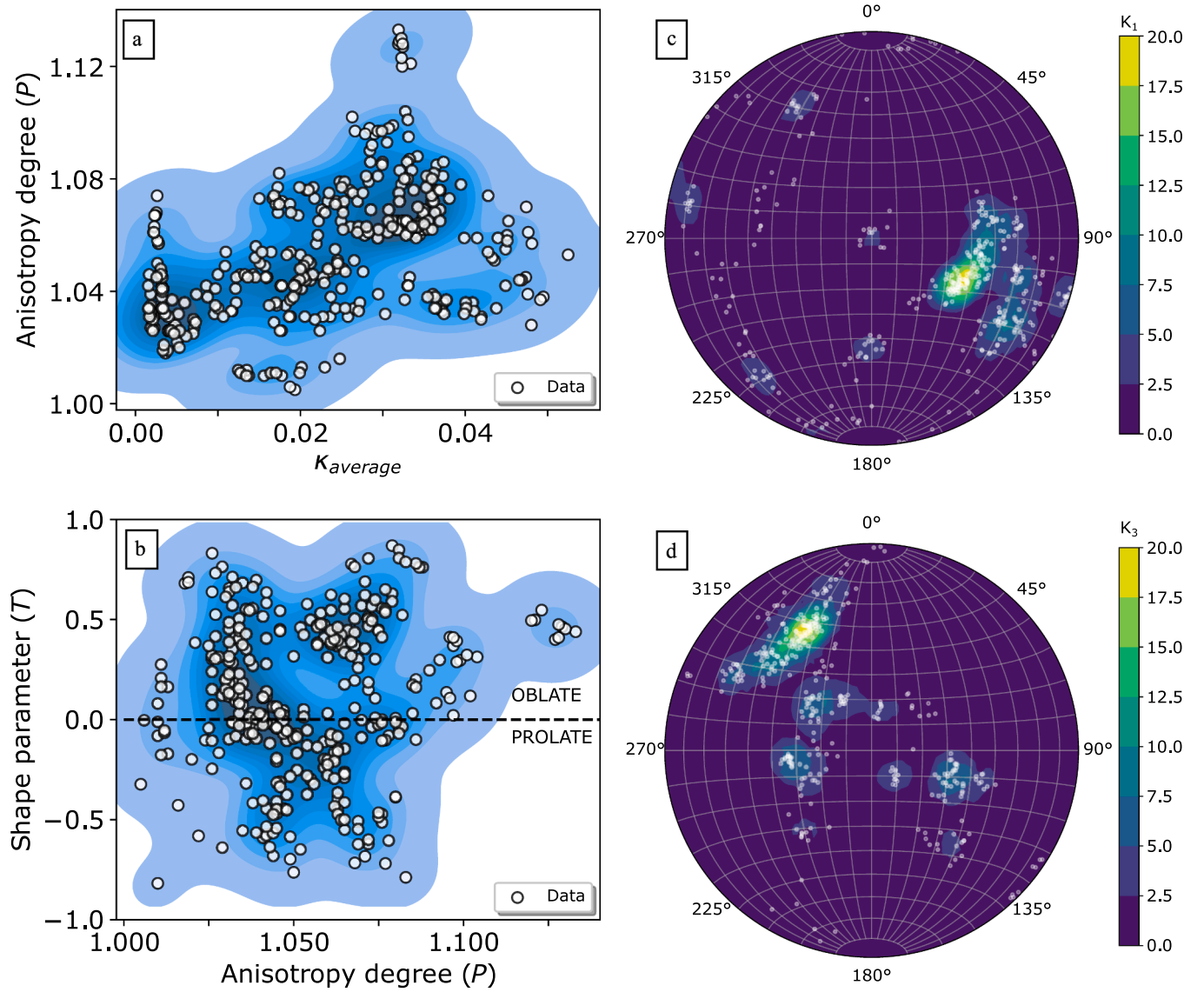


Fig. 5. AMS data of individual specimens of the Rio Perdido samples. (a) Bulk magnetic susceptibility ($k_{average}$) × anisotropy degree (P) shows no evident correlation, as well as the (b) Jelínek plot, where ellipsoid shapes are distributed between oblate and prolate ones. (c and d) Stereographic distributions of maximum (K_1) and minimum (K_3) susceptibility directions, respectively. Density contour units are counts in 1 % of area.

4.2.1. Baked contact test

Six samples collected in the host rock up to a meter away from a 2 m wide dike on site 22 (samples SR32, Table 1) show same SW directions with low inclination, similar to that from the dike, while two samples in the host rock collected 3.6 and 6.10 m from the dike yielded NW directions with moderate (to steep) upward inclination (Fig. 9). Most baked samples (<1m distant from contact) presented only SW directions becoming unstable at coercivities above 20–25 mT (see Supplementary-File 3). The exception was sample SR32L (sampled at 0.39 m from contact), for which the AF treatment isolated two stable directions: the SW direction with low inclination, carried by coercivities between 4 and 18 mT and the NW basement direction, with moderate negative inclination, carried by higher coercivities, between 40 and 100 mT (see Fig. S1g of the Supplementary-File 3). Magnetic mineralogy study of this sample revealed magnetic properties similar to the own dike, although it presented a weaker magnetic momentum, resulting in noisier data (Fig. 10). The magnetic hysteresis loop (Fig. 10a) indicates a low coercivity and a small paramagnetic contribution. Ratios such as M_{rs}/M_s (0.186) and B_{cr}/B_c (2.39) fall within the pseudo-single domain (PSD) field of the Day plot (Day et al., 1977; Dunlop, 2002). The isothermal remanent magnetization curve (IRM, Fig. 10b) reveals that the majority of magnetization is acquired before reaching 200 mT, with saturation practically achieved at 400 mT. In the FORC diagram (Fig. 10c), the coercivity axis primarily shows a distribution between 5 and 25 mT, but there is also a notable spreading toward the Bu-axis. While this spreading aligns with PSD behavior (Roberts et al., 2000, 2022), similar to what is observed in the dikes (see above), we interpret the larger spreading along the Bc-axis as likely associated with a population of magnetite exhibiting variable grain sizes within the PSD domain.

Isolated directions for the dike, baked samples and unbaked samples are shown in Fig. 9a', 9b' and 9c', respectively, which yielded the following mean directions (see also Table 1): dike mean direction – Dec = 219.7°; Inc = -19.4° (α_{95} = 8.4°; k = 64.0); baked mean direction – Dec = 199.4°; Inc = 4.4° (α_{95} = 11.0°; k = 19.4); unbaked mean direction – Dec = 305.4°; Inc = -39.5° (α_{95} = 8.1°; k = 55.9).

Therefore, from the above, we interpret these results as a positive baked contact test, which attests that AF and thermal demagnetization successfully isolated the thermoremanent magnetization acquired during dikes' intrusions.

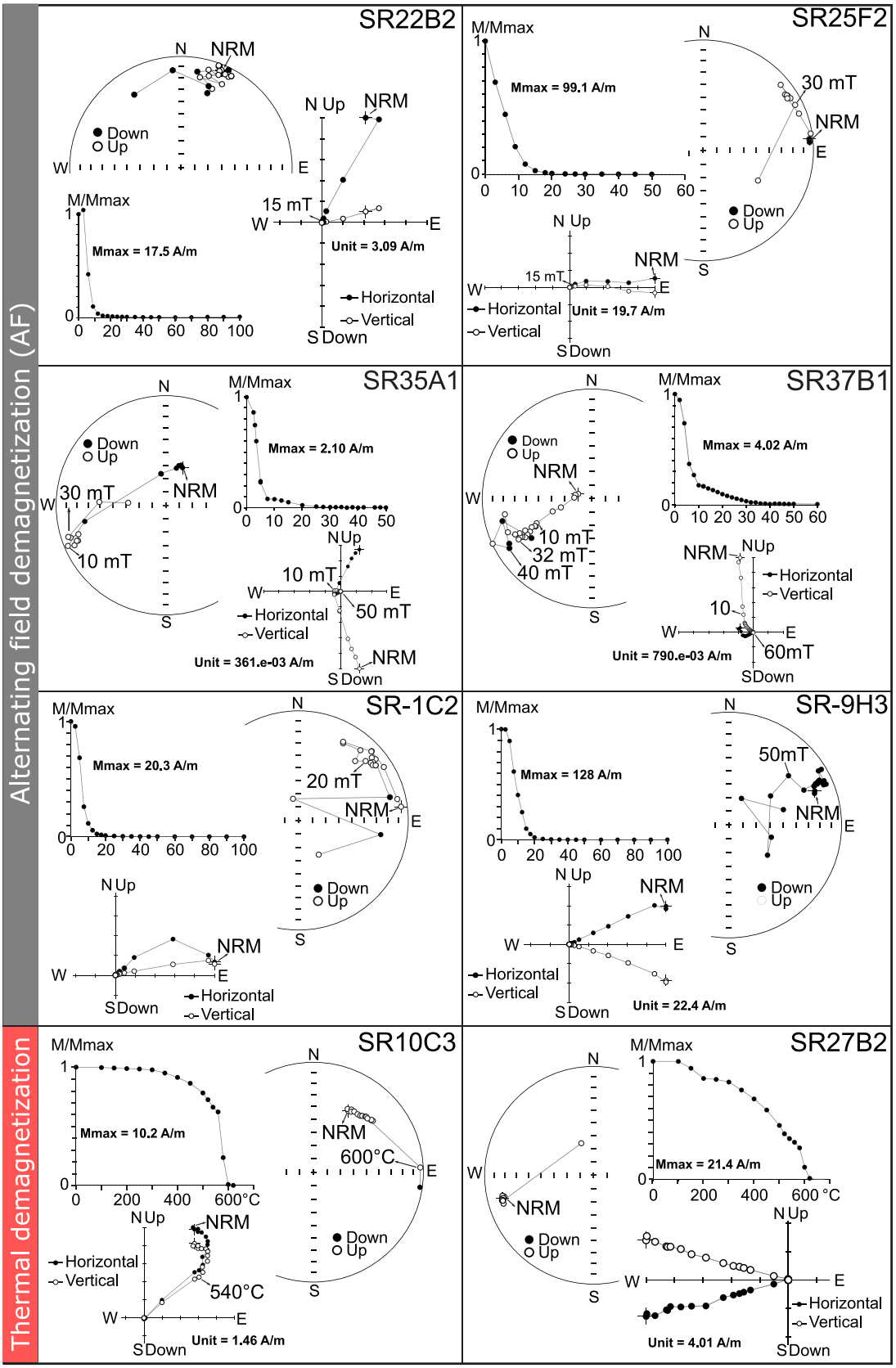


Fig. 6. Examples of alternating field and thermal demagnetization procedures for samples of the Rio Perdido dikes. For each sample, the components are shown in an orthogonal (Zijderveld) projection, and their vectorial trajectory is reflected in a stereographic projection, followed by their normalized (M/Mmax) magnetization curves versus the alternating field (mT)/temperature (°C).

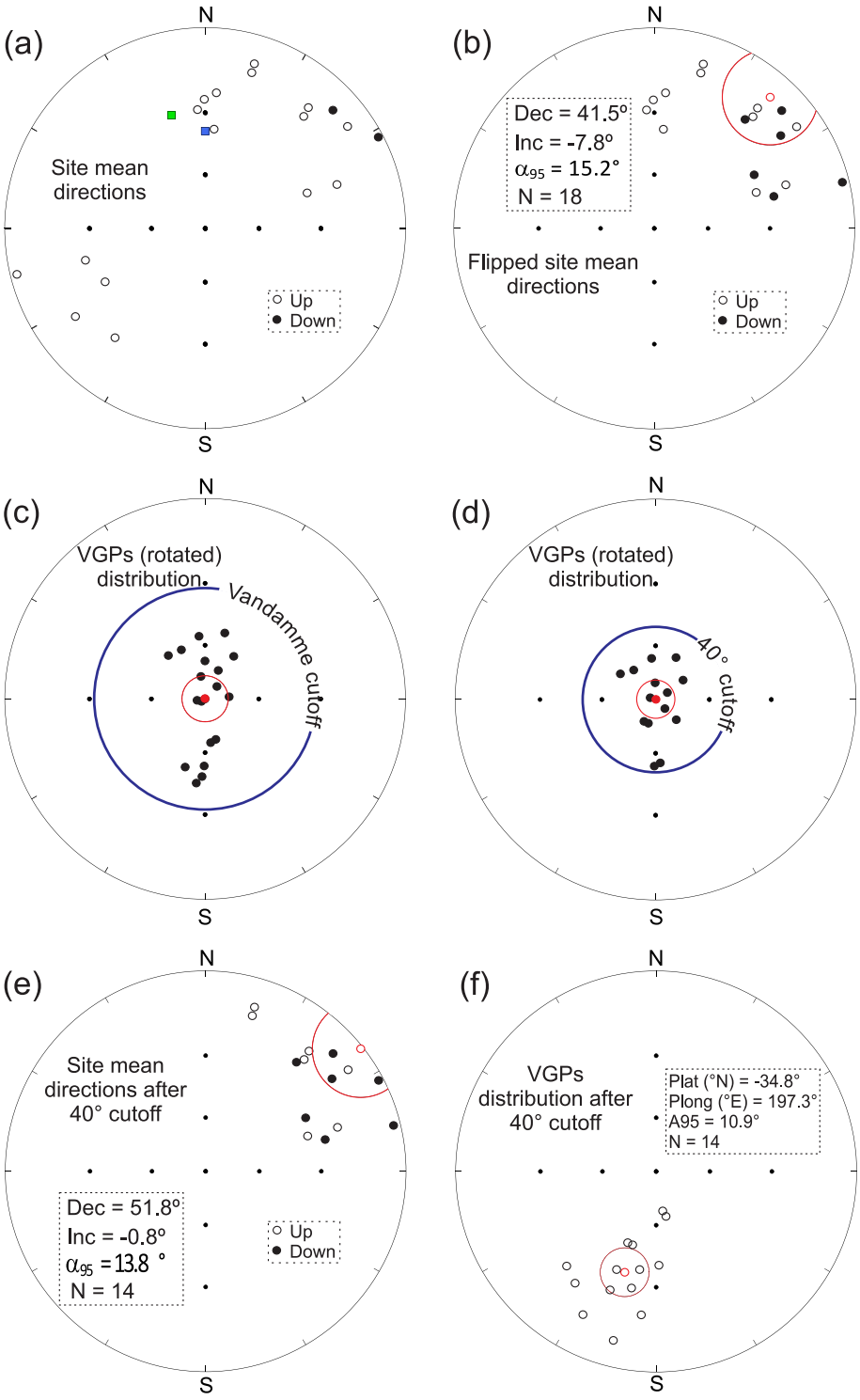


Fig. 7. a) Site mean directions (N = 18) for the Rio Perdido dikes. Green and blue squares are the present geomagnetic and dipolar fields, respectively. b) The southwestern directions were flipped and the calculated mean magnetization direction is shown in red color with the respective confidence cone (α_{95}) c) Vandamme's cutoff was applied to VGPs distribution, but no VPGs were excluded with such a procedure. d) Instead, a 40° cutoff allowed us to remove 4 VPGs to produce a new mean site cluster (e) and, consequently, a paleomagnetic pole with a smaller confidence cone (f). Inner red circles represent either the site mean direction or the mean VPG, as the outer red circles are their α_{95}/A95 confidence interval. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. Discussion

5.1. Paleomagnetic pole reliability

The new Rio Perdido (RP) paleomagnetic pole (34.8°S; 197.3°E) satisfies six out of seven of the Van der Voo (1990) and Meert et al. (2020) reliability criteria, as follows: (1) U–Pb ages (ID–TIMS) dating on four baddeleyite fractions of Rio Perdido dikes yielded an age of 1110.1 ± 2.3 Ma (Teixeira et al., 2019), which is interpreted as the crystallization age of this suite. Although this dike was not sampled for our

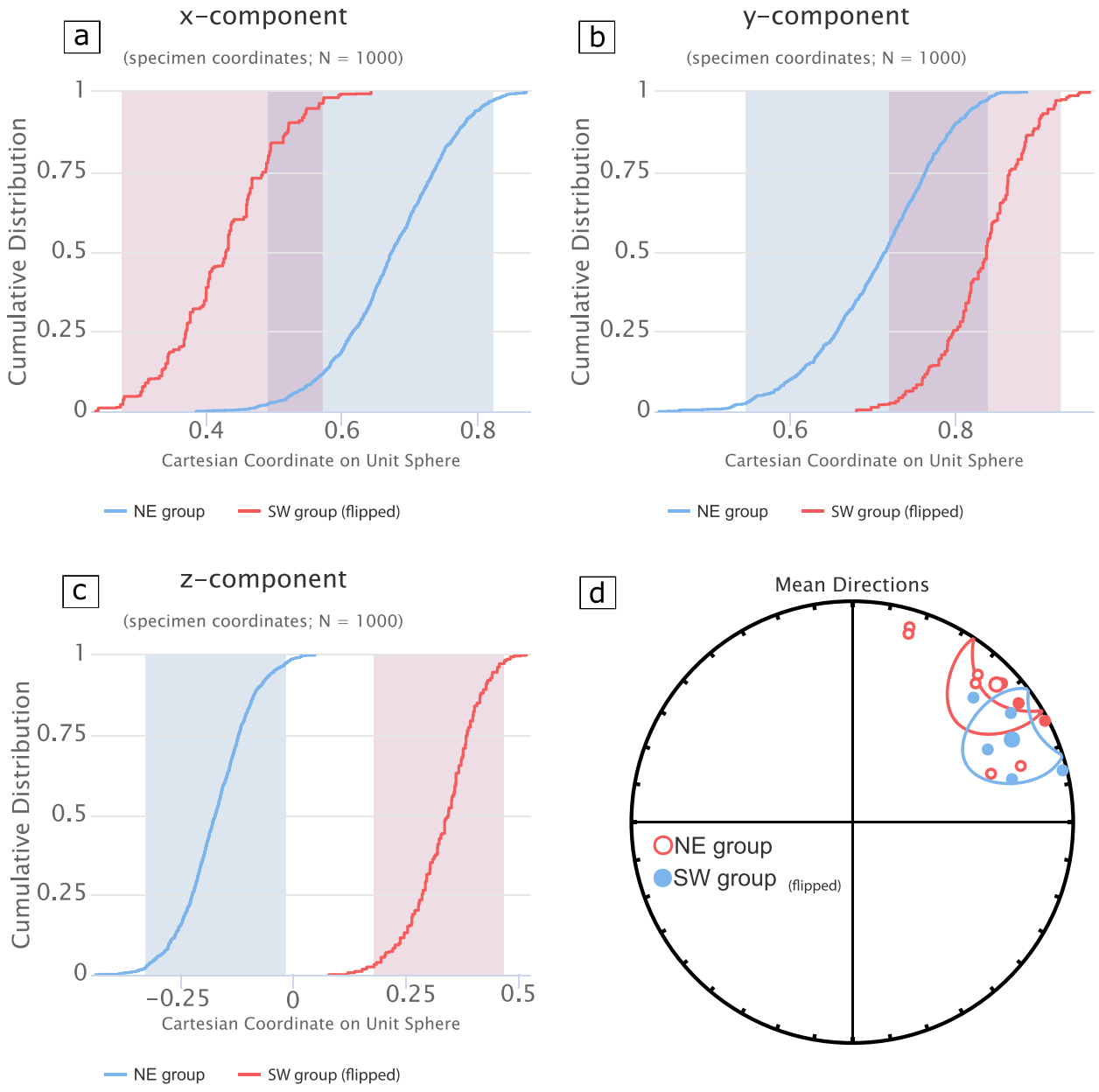


Fig. 8. True Mean Directions (Tauxe, 2010) test for site direction means with opposite polarities (NE and SW groups) performed at paleomagnetism.org (Koymans et al., 2016). There is overlapping for the × (a) and y (b) components, but not for the z-component. Therefore, although the α_{95} of both groups intersect each other (d), the directions are too dispersed to be considered the same with 95 % confidence.

paleomagnetic study, it belongs to the same swarm.; (2) Stepwise AF and thermal demagnetization procedures were applied to achieve the vectorial data, which were sequentially used to calculate a ChRM component from principal component analysis (PCA) from orthogonal diagrams (Kirschvink, 1980). A total of 129 samples from 14 sites were used in the calculation of the RP pole, which produced the Fisher's statistic parameters (A95 = 10.9°, K = 14.3) which meet the R2 criterion of Meert et al. (2020), that is, N ≥ 25; 10 ≤ K ≤ 70 and B ≥ 8. Also, this pole passes Deenen's criteria for paleosecular variation (see above); (3) Thermomagnetic, hysteresis and IRM curves and FORC diagrams indicate that PSD Ti-poor titanomagnetite is the main magnetic carrier of the remanent magnetization present in these rocks; (4) A positive backed contact test was achieved for site 22 (Table 1), attesting the isolated characteristic magnetization as a primary thermoremanence and, by consequence, dated as stated in the criterion (1); (5) The Rio Perdido dikes are the last tectonic event that occurred in the RAT, so that it maintained its integrity after dike intrusions. Also, the dikes do not show petrographic evidence of metamorphism and/or deformation (Lima, 2016; Lima et al., 2017). Furthermore, AMS data indicates low anisotropy degrees and a magnetic foliation following their general trend, which is coherent with an undeformed magmatic fabric; (7) Although the RP pole is the first determined for RAT, it does not coincide with any younger pole for South America. Also, the positive baked contact test implies that no later remagnetization event affected the studied rocks.

On the other hand, criterion 6 is not fully obeyed. Although the Rio Perdido ChRMs record two polarities (Fig. 7), the classical reversal test of McFadden and McElhinny (1990) and the CTMD test conducted at paleomagnetism.org yielded negative results at a 95 % confidence level (see discussion above).

Nevertheless, the 1110 Ma Rio Perdido pole can be considered a key pole for the RAT during the Mesoproterozoic, and its paleogeography relative to the Amazonian Craton and other cratonic blocks is discussed below.

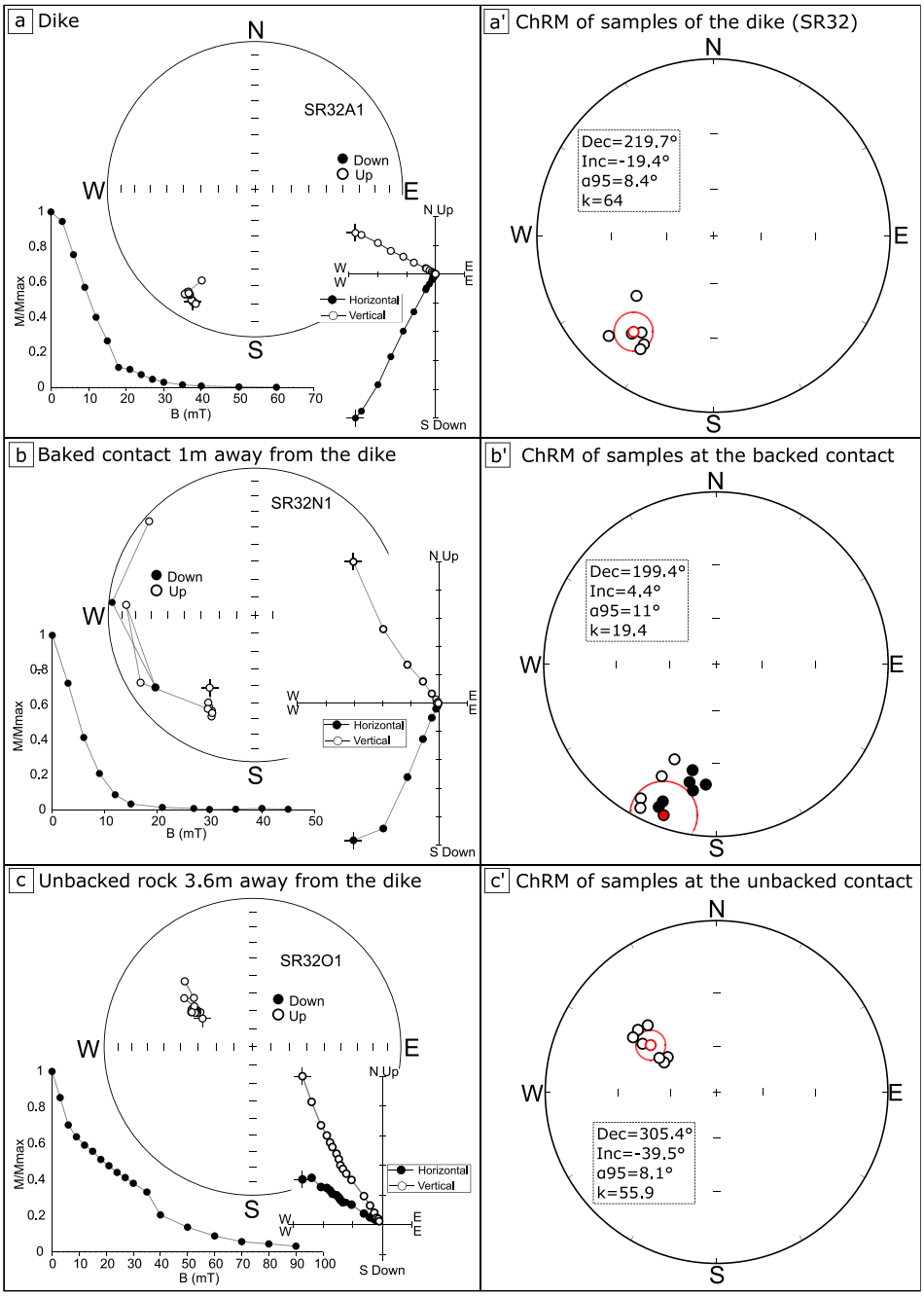
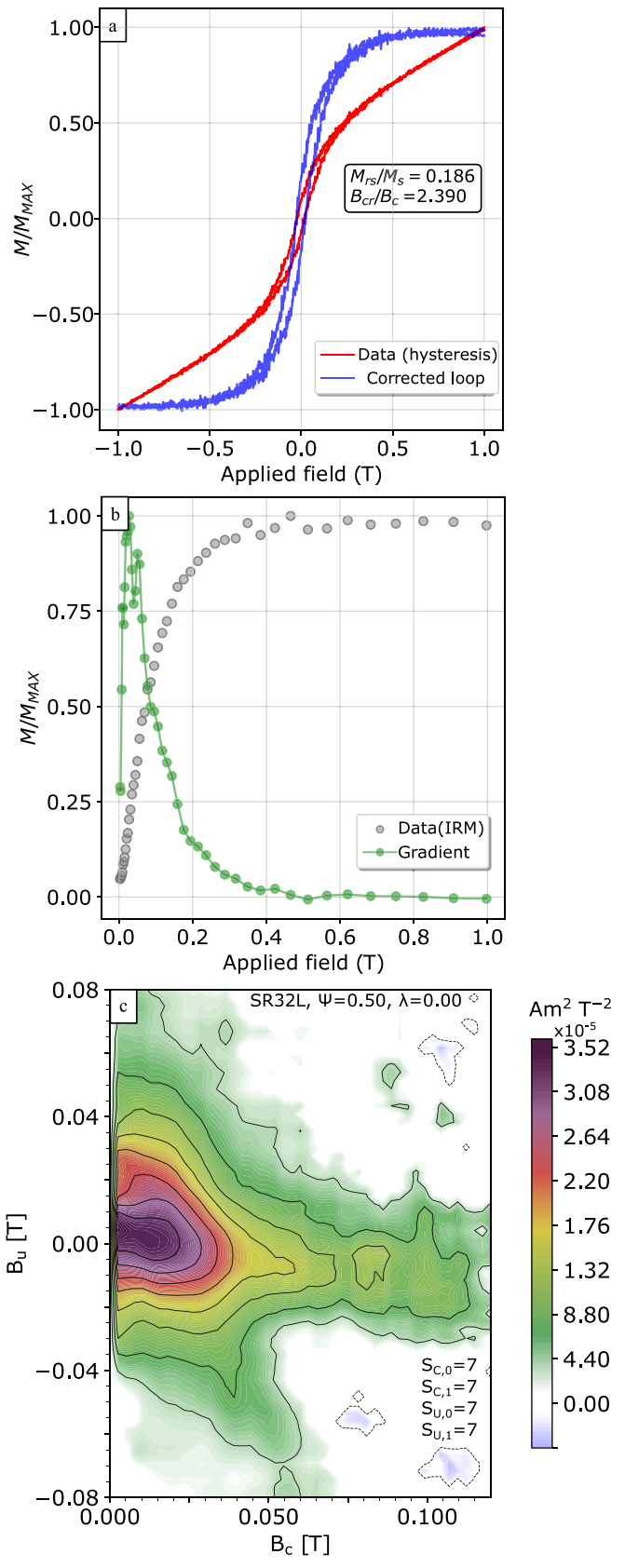


Fig. 9. a, b. c) Backed contact test of samples from site 22 (samples SR32). Alternating field demagnetization procedure shows similar SW directions with low inclinations for the dike and a baked host rock one meter from it. Gneissic rocks at 0.39 m, 3.6 m and 6.1 m away from the dike's contact show distinct NW directions with moderate to steep inclinations (The Supplementary-File 3 shows stereographic projections of AF demagnetization for samples from the dike and gneissic rocks at different distances from contact). a', b', c') Isolated directions for samples from the dike, from the baked gneissic samples and for the unbaked gneissic samples, respectively. Mean directions and their respective confidence cone (α_{95}) are shown in red. These results attest that the RP pole results from a primary thermo-remanent magnetization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.2. *Testing the Amazonia and Rio Apa connection*

The role of the RAT relative to the Amazonian Craton (AC) during Precambrian times has long been discussed in the literature (e.g., Cordani et al., 1979; Litherland et al., 1986; Ruiz et al., 2005; Cordani et al., 2010; Redes et al., 2015; 2020; Lacerda-Filho et al., 2016; Faleiros et al., 2016; Ribeiro, et al., 2020; 2022a; 2022b; Nedel et al., 2020; Teixeira et al., 2020). Currently, RAT is separated from AC by the Brasiliano Tucavaca Belt (Fig. 1), which was interpreted as an aulacogen belt by Ruiz et al. (2005) and Cordani et al. (2010). Also, geological, geochronological and zircon U–Pb–Hf isotopic data strongly suggest a coeval Paleo- to Mesoproterozoic evolution of the RAT and the Paraguá Terrane (PT), presently in the southwestern AC (e.g., Redes et al., 2015; 2020; Nedel et al., 2020; Ribeiro et al., 2022a; 2022b). Ribeiro et al. (2022a; 2022b) suggest that RAT and PT behaved as a single evolutionary continental block that collided with AC at ca. 1320 Ma, along the Alto Guaporé Belt (e.g., Rizzotto and Hartmann, 2012; Rizzotto et al., 2014). This is corroborated with the fact that at 1100 Ma the Rio Perdido dikes from RAT and the coeval Huanchaca sills and Rincón del Tigre mafic layered intrusive Suite from AC formed a large igneous province (LIP) (Teixeira et al., 2020).

Recently, Bispo-Santos et al. (2023) published paleomagnetic data



(caption on next column)

Fig. 10. Magnetic mineralogy data for the gneissic sample SR32L collected at 0.39 m from dike's contact: a) Magnetic hysteresis, displaying the acquired data (in red) alongside the dia/paramagnetic correction. b) Isothermal remanent magnetization (IRM) curve (in gray) and its gradient (in green). c) FORC diagram showing PSD behavior (Roberts et al., 2000, 2022), similar to what is observed in the dykes (see above). We interpret the larger spreading along the Bc-axis as likely associated with a population of magnetite exhibiting variable grain sizes within the PSD domain. Parameters illustrated in the First Order Reversal Curve (FORC) diagram are thoroughly discussed in the topic 3 (Sampling and Procedures) - As depicted, higher smoothing factors were deemed necessary for the FORC diagram due to the considerably weaker magnetization in comparison to the dykes, which resulted in a lower signal-to-noise ratio. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

for the 1110 Ma Huanchaca sills from the Amazonian Craton. As a result of their study, the mean direction Dm = 302.2°; Im = 12.2°, (K = 12.0; α_{95} = 14.5°), and the paleomagnetic pole (HU) situated at 30.1 °N; 225.7 °E (A95 = 9.9°) were calculated for 10 analyzed sites collected from the Huanchaca sills (Table 2). Therefore, the paleomagnetic pole (RP - Table 1) obtained for the 1110 Ma Rio Perdido dikes in this study allows testing the relative position of the RAT regarding the AC during the late Mesoproterozoic. The low mean inclinations obtained for both Huanchaca (Im = 12.2°; paleolatitude = 6.2°) and Rio Perdido (Im = -0.8; paleolatitude = -0.4°) igneous suites indicate that AC and RAT were close to the Earth's Equator at 1110 Ma. However, the mean declination of the Rio Perdido dikes (Dm = 51.8°) is very different from that obtained for the Huanchaca sills (Dm = 302.2°), which indicates that RAT was rotated ~ 110° relative to AC (compared with their present relative positions).

A possible paleogeographic scenario for the RAT and AC at 1110 Ma is shown in Fig. 11a. The southwestern AC and RAT have been enlarged in Fig. 11b, and the main geological/geotectonic features were included, according to Teixeira et al. (2020). Unfortunately, most of the RAT is covered by late Neoproterozoic to Phanerozoic sedimentary rocks, but geochronological data of the Taquaral and Coimbra batholiths (Redes et al., 2015; Faleiros et al., 2016; Santos et al., 2019) of the Corumbá Window (CW in Fig. 11b) indicates that these rocks are probably an extension to the north of the Western Domain (WD in Fig. 11b) of the RAT (Teixeira et al., 2020). Furthermore, several granitic bodies such as Las Torras (Nedel et al., 2021), Correreca (Vargas-Mattos et al., 2010), Santo Corazón (Redes et al., 2020) and Santa Terezita (Redes et al., 2018)) and a granodiorite batholith (Nedel et al., 2021) located to the south of the San Diablo shear zone (see Fig. 11b) yielded ages between 1940 and 1860 Ma, suggesting this region can also be part of the same geological evolution of the Corumbá Window and the Western Domain of the RAT (Nedel et al., 2021), although additional zircon U–Pb–Hf is required to confirm such hypothesis.

The San Diablo shear zone developed at ca. 1000 Ma along older structures and weakness zones (Nedel et al., 2021). The authors suggest that it separated the older (1900–1800 Ma) southern domain from the 1700–1600 Ma northern domain (so-called Paraguá Terrane - PT), whose collision between them would have happened at ca. 1350 Ma, during the San Ignacio Orogeny (see also Ribeiro et al., 2022a, 2022b). The RAT was also largely affected by this event (e.g., Cordani et al., 2010; Lacerda-Filho et al., 2016; Ribeiro et al., 2020; 2022a; 2022b). However, it is a consensus among these authors that the basement and the magmatic arcs of the RAT were developed and stable at ca. 1670 Ma. Here, we envisage that the RAT, SW Amazonia and PT evolved together as a single block and collided with the southwestern AC at ca. 1320 Ma along the Alto Guaporé Belt (e.g., Bettencourt et al., 2010; Rizzotto and Hartmann, 2012; Rizzotto et al., 2014). Also, the paleogeographic configuration shown in Fig. 11 at 1110 Ma ago is supported by paleomagnetic data.

Table 2
Selected ca. 1,100 Ma paleomagnetic poles for Baltica, Laurentia, Amazonia, Rio Apa, West Africa, Kalahari, Congo-São Francisco and India and Euler rotation poles for the reconstruction of Figs. 12 and 13a.

CRATON / Geological unit	Code	Age Ma	PLat. (°N)	PLon. (°E)	A95(°)	Rlat. (°N)	Rlong. (°E)	R	Ref.
BALTICA – Rotation pole: 39.53°N, 77.01°E (45.6°)									
Salla Diabase Dyke	SD	1122 ± 7	71.0	113.0	8.1	71.8	280.1	5	1
LAURENTIA – Rotation pole: 3.80°N, 143.6°E (41.46°)									
Lower Mamainse Point volcanic – R2	L1	1105 ± 2	37.5	205.2	4.5	67.6	164.7	7	2
Upper Osler volcanic-R	L2	1105 ± 2	42.5	201.6	3.7	67.9	149.6	7	2
Middle Osler volcanic-R	L3	1107 ± 3	42.7	211.3	8.2	74.3	159.5	6	2
Lowermost Mamainse Point volcanic s-R1	L4	1108 ± 3	49.5	227.0	5.3	86.5	128.8	7	2
Lower Osler volcanic-R	L5	1108 ± 3	40.9	218.6	4.8	77.7	178.9	6	2
Mean Nipigon sills and dykes	L6	1111 ± 4	47.2	217.8	4.0	80.3	148.1	5	2
AMAZONIA - Rotation pole: 48.49°N, 176.73°E (95.7°)									
Huanchaca sills	HU	1112 ± 2	30.1	225.7	9.9	88.4	256.2	5	3
WEST AFRICA – Rotation pole: 12.43°N, 194.21°E (101.74°)									
CONGO - Rotation pole: 13.81°N, 195.64°E (126.43°)									
Huila Epembe	HE	1110 ± 3; 1109 ± 10	−34.7	256.5	8.7	84.5	276.3	6	4
SÃO FRANCISCO - Rotation pole: 41.04°N, 174.22°E (117.58°)									
INDIA - Rotation pole: 52.56°N, 281.14°E (100.26°)									
Mahoba dykes	MH	1113 ± 7	+38.7	229.5	12.4	84.0	10.0	4	5
KALAHARI - Rotation pole: 14.73°N, 196.54°E (169.56°)									
Post-Gupera dykes	PG	1105 ± 1	62.3	31.9	6.9	85.9	327.1		2
Umkondo mean	UK	1109 ± 1; 1112 ± 1	−64.0	222.1	2.6	84.1	22.3	5	6
RIO APA – rotation pole: 44.23°N, 317.15°E (81.969°)									
Rio Perdido dykes	RP	1110 ± 2	34.8	17.3	10.9	88.6	206.1	7	7

Plat. – Pole latitude; Plong. – Pole longitude; A95 – radious of 95 % confidence cone; Rlat – rotated pole latitude; Rlong – rotated pole longitude; Reference: 1- Salminen et al. (2009); 2 - Evans (2021); 3 – Bispo-Santos et al. (2023); 4- Salminen et al. (2018); 5 - Pradhan et al. (2012); 6 - Swanson-Hysell et al. (2015); 7- This work. For reconstruction of Fig. 13b, Umkondia and respective poles (Rlat., Rlong.) were rotated 30° clockwise around the Euler pole at (90°N, 0°E).

5.3. *Where did the Rio Apa/Paraguá Terrane come from?*

The geochronological and zircon U–Pb–Hf isotopic data for rocks from the RAT, PT and Ventuari–Tapajós/Rio Negro–Juruena provinces imply a similar geological evolution of these terranes during Orosirian and Statherian, suggesting they were close together at that time (e.g., Ribeiro et al., 2020; 2022a; 2022b; Teixeira et al., 2020).

A baked contact test was performed in this study for a dike cross-cutting gneissic rocks of the Caracol magmatic arc. This test returned positive, since samples collected close (up to 1 m from contact) to the dike yielded directions similar to that from the own dike (Fig. 9), but a very different stable remanent magnetization was isolated for samples collected at 3.6 m (SR32-O) and 6.1 m (SR32P) away from the dike contact and sample SR32L located at 0.39 cm from the dike yielded both the dike direction and the basement characteristic remanent direction (see Fig. 9). A mean direction was calculated for these (Dm = 305.4°, Im = -39.5°, N = 7, α_{95} = 8.1°, k = 55.9) which yielded a virtual geomagnetic pole (VGP) at 39.6°N, 200.9°E (dp = 5.8°, dm = 9.7°). Zircon U–Pb ages from the Caracol rocks range from 1783 ± 18 Ma to 1721 ± 25 Ma with most crystallization ages clustering at 1770–1760 Ma (Teixeira et al., 2020; Ribeiro et al., 2020). Although this VGP (designated as CG) does not represent the Earth's geographic pole at 1770–1760 Ma, since it is affected by geomagnetic secular variation, it can be used to estimate the position of the RAT (and associated PT) relative to the proto-Amazonian Craton (proto-AC) at that time. For that, we may compare CG (VGP) with the slightly older 1780–1790 Ma Avanavero paleomagnetic pole for the proto-AC, located at 48.4°N, 207.9°E (A95 = 9.9°) (Bispo-Santos et al., 2014). A possible paleogeographic configuration is shown in Fig. 12a, where the PT is next to the proto-AC. Note that the RAT position relative to PT is the same as observed today in South America. A possible scenario is to consider that the RAT/PT block separated from proto-AC after 1.76 Ga and collided again with proto-AC along the Alto Guaporé Belt at ca. 1340–1320 Ma (Bettencourt et al., 2010; Rizzotto and Hartmann, 2012; Rizzotto et al., 2014). Although a speculative model, a detailed study of the Caracol gneissic rocks may corroborate it in further studies.

5.4. *The RAT in Rodinia*

Based on paleomagnetic data, Bispo-Santos et al. (2023) tested the position of the Amazonian Craton relative to other cratonic blocks at 1110 Ma. These authors followed Choudhary et al. (2019) who proposed the megacontinent Umkondia formed by Amazonia, West Africa, Kalahari, Congo-São Francisco, and India. We reproduce this model in Fig. 12b, with the inclusion of RAT, based on the 1110 Ma Rio Perdido pole (as in Fig. 11). In the model of Choudhary et al. (2019), Umkondia would have been formed through a transcurrent strike-slip motion of the Kalahari–Congo–São Francisco–India relative to Amazonia–West Africa after ca. 1380 Ma. We suggest that during this (sinistral) movement, the RAT rotated ~ 110° counterclockwise relative to PT/Amazonia, as suggested by the 1110 Ma Rio Perdido and Huanchaca poles (see Fig. 11). Also, Choudhary et al. (2019) suggest that (i) the ca. 1110 Ma Huila-Epembe dikes and related mafic rocks in the Congo Craton (Salminen et al., 2018), (ii) the 1110 Ma Umkondo LIP in the Kalahari Craton (Swanson-Hysell et al., 2015), (iii) the 1110 Ma Rincón del Tigre-Huanchaca intrusions in the Amazonian Craton (Teixeira et al., 2015), (iv) the 1110 Ma Rio Perdido dikes in the RAT (Teixeira et al., 2019) and (v) the 1113 Ma Mahoba dikes in India (Pradhan et al., 2012) originated from the same plume center situated at the present northwestern Kalahari Craton (red star in Fig. 12b). Note that the present-day WNW-ESE

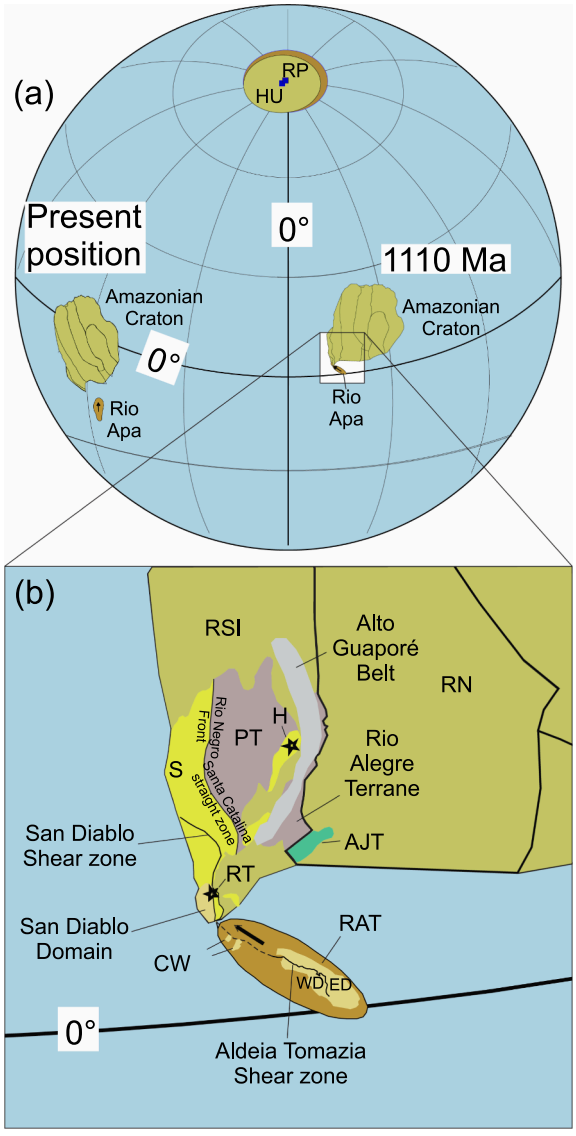


Fig. 11. (a) Possible paleography of RAT relative Amazonia at 1110 Ma. Euler rotation poles used in this reconstruction for Amazonia and RAT are 48.49°N, 176.73°E 95.7°) and 44.23°N, 317.15°E (81.969°), respectively. RP- Rio Pedido pole; HU- Huanchaca pole (Table 2). The present position of RAT and Amazonia is also shown for comparison. (b) Expanded NW Amazonia and RAT shown their main geological/tectonic features. RN – Rio Negro-Juruena Province; RSI – Rondonian-San Ignacio Province; AJT – Alto Jauru Terrane; PT – Paraguá Terrane; S – Sunsás Province; H – Huanchaca sills; RT – Rincón del Tigre Intrusive suite; CW – Corumbá Window; WD – Western Domain; ED – Eastern Domain.

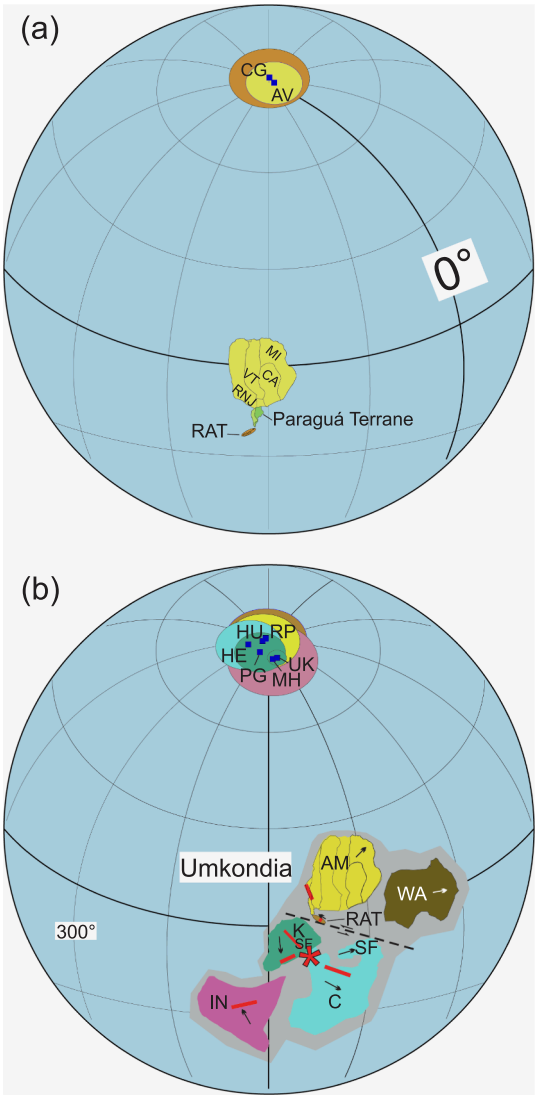


Fig. 12. (a) Possible paleogeography of proto-Amazonia and Paraguá Terrane/RAT at 1780–1770 Ma. Euler rotation poles used in this reconstruction for proto-Amazonia and Paraguá/RAT are 0°N, 297.9°E (-41.6°) and 13.439°N, 117.443°E (53.399°), respectively. CA – Central Amazonia Province; MI – Maroni-Itacaiunas Province; VT – Ventuari-Tapajós Province; RN – Rio Negro-Juruena Province. (b) Test of the Umkondia megacontinent (Choudhary et al., 2019) with the inclusion of RAT (modified after Bispo-Santos et al., 2023). Cratons: AM – Amazonia; WA – West African; K – Kalahari; SF – São Francisco; C – Congo; IN – India. Arrows indicate the present North of each craton. Paleomagnetic poles (with symbols) and Euler rotation poles in Table 2.

Rio Perdido dikes strike in the SW-NE direction in Umkondia, which is coherent with the plume center at the northwestern Kalahari (see Fig. 12b). The 1110 Ma paleomagnetic poles from each of these cratonic units cluster around the north geographic pole when we consider the Umkondia reconstruction (Fig. 12b; poles and Euler rotation poles are available in Table 2).

Bispo-Santos et al. (2023) proposed that Umkondia rotated clockwise after 1110 Ma colliding with Laurentia and Baltica to form the Rodinia supercontinent at ca. 1000–900 Ma. These authors proposed two paleomagnetically possible models which are reproduced in Fig. 13. In the first model, the Amazonia–West Africa was united to Baltica and Laurentia as in the configuration of Columbia (Fig. 13a). Then, Baltica (linked to Umkondia) broke-up from Laurentia after ca. 1110 Ma and rotated clockwise until it docked again with Laurentia at 1000–900 Ma.

Similar models considering clockwise rotations of Baltica and Amazonia colliding with Laurentia to form Rodinia were previously proposed by several authors (Sadowski and Bettencourt, 1996; Johansson, 2009; Cawood and Pisarevsky, 2017; Martin et al., 2020; Antonio et al., 2021; Johansson et al., 2022, among others), although the time when the Amazonia and Baltica broke away from Columbia and collided again with Laurentia to form Rodinia is still controversial. For example, Johansson (2009) and Johansson et al. (2022) suggest the rupture of Baltica–Amazonia from Laurentia would have occurred at ca. 1260 Ma. However, the rupture of Baltica from Laurentia may also have occurred after 1110 Ma with the formation of the Asgard sea (Cawood and Pisarevsky, 2017).

In Fig. 13b, Umkondia appears separated from Baltica–Laurentia at ca. 1110 Ma. Then, Baltica broke up from Laurentia and rotated clockwise while Umkondia did the same colliding with Laurentia and Baltica

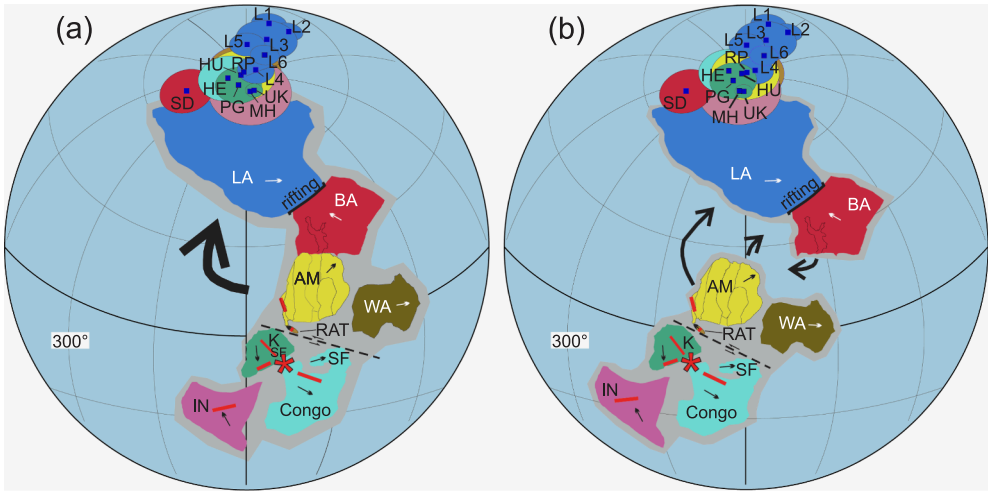


Fig. 13. (a) Paleogeographic reconstruction of Umkondia, Baltica and Laurentia at 1110 Ma in the Rodinia supercontinent. (b) The same as in (a), but with umkondia rotated 30° clockwise around the North geographic pole. Modified after Bispo-Santos et al. (2023). BA – Baltica; LA – Laurentia; Symbols in the other cratons as Fig. 12. Paleomagnetic poles (symbols) and Euler rotation poles are presented in Table 2.

at ca. 1000–900 Ma. The main reason argued by Bispo-Santos et al. (2023) to separate Umkondia from Baltica–Laurentia is the development of the ca. 1000 Ma Putumayo orogeny (Ibañez-Mejia et al., 2011; Ibañez-Mejia, 2020) in northwestern Amazonia (see also Cawood and Pisarevsky, 2017). Recently, a similar model was also proposed by Martin et al. (2022) who consider that Amazonia separated from the Columbia supercontinent at ca. 1380 Ma and Baltica at ca. 1110 Ga, both executing a clockwise rotation until they collided again with Laurentia at 1000 Ma.

5.5. RAT in the formation of Gondwana

The West-Gondwana was formed at ca. 600–520 Ma ago with the collision of two continental masses along the Transbrasiliano-Kandi tectonic corridor (TB-K in the inset of Fig. 14a) linking the West Africa/Amazonian Craton (linked to RAT) with the Sahara Metacraton/São Francisco/Congo/Rio de la Plata/Kalahari and other eventual hidden cratonic block in the Paraná basin, named as Paranapanema craton (Cordani et al., 2013; Ganade de Araújo et al., 2014; Trindade et al., 2006). The Paraguay-Araguaia mobile belt separates the Amazonian (including the RAT) Craton from the São Francisco and Paranapanema cratons. The Paraguay belt surrounds the RAT forming a concave curvature from the south to a convex curvature surrounding the Amazonian Craton to the north (Fig. 14a). During the formation of West Gondwana we can envisage that RAT probably rotated ~ 110° relative to a growing Amazonia during collision resulting in the current geographic position of RAT distal to the Amazonian craton.

Paleomagnetic studies of Ediacaran limestones and dolomites (specifically from the the Araras Group and Bocaina Formation – Fig. 14b and 14c) situated on both sides from the oroclinal bending (E–W trend in the north, changing to N-S trend in the south) of the Paraguay Belt revealed ~ 90° difference in magnetic declinations (Fig. 14 d), likely due to collisional events (Tohver et al., 2010). These authors estimate an age of ca. 528 Ma for the time magnetic directions were acquired by the rocks, based on Ar–Ar dating, considered as the maximum age for the rock authigenesis/remagnetization. North-northeastern, downward inclination directions were found for the N–S trend of the oroclinal (surrounding the RAT – Fig. 14), while southeastern, downward directions were found for the NNW–SSE trend segment (surrounding the Amazonian Craton). Interestingly, the northeastern directions agree with those disclosed for the ca. 525 Ma Itabaiana dikes from the Borborema Province (Fig. 14d, Trindade et al., 2006), for Neoproterozoic carbonates from the Sete Lagoas Formation (D'Agrella-Filho et al., 2000) and Salitre Formation (Trindade et al., 2004), both from the São Francisco Craton, and whose characteristic remanent magnetization was interpreted as due to remagnetization events occurred at ca. 525 Ma. At first, this rotation of 90° in declination along the oroclinal bending could be related to the clockwise rotation of RAT after ca. 525 Ma. However, some evidence contradicts this association: (i) the northeastern declinations (not affected by a possible rotation) were found for the south segment of the oroclinal surrounding RAT; (ii) we could envisage a 90° counterclockwise rotation of the Amazonia relative to RAT which would explain the southeastern declinations found for carbonates in the northern sector of the oroclinal bending. However, the same north-northeastern declinations with downward inclinations were found for Puga cap carbonates (from the Araras Group) in the Amazonian Craton, which were interpreted as result of a process of rock remagnetization occurred at ca. 525 Ma ago (Trindade et al., 2003), suggesting that the 90° variation in declination along the Paraguay oroclinal bending is most probably due to a local tectonic event. Thus, the ~ 110° rotation of RAT relative to the Amazonian Craton during the amalgamation of Gondwana, as suggested by the Rio Perdido and Huanchaca paleomagnetic data, would have to have happened before ca. 525 Ma.

6. Conclusion

A mean direction Dec = 51.8°, Inc = -0.8° (α_{95} = 13.8°, k = 9.3) was calculated for 14 dikes from the 1110 Ma Rio Perdido dike swarm (RAT), which yielded the paleomagnetic pole (RP) at 34.8 °S/197.3 °E (A95 = 10.9°). This pole passed a baked contact test and is classified with R = 6 in the reliability criteria of Meert et al. (2020). Based on this new paleomagnetic pole gathered with other poles available for a similar time (ca. 1110 Ma) permitted to test the position of RAT in paleogeographic reconstructions based on geologic, geochronologic and isotopic data. Our study demonstrated that the inferred connection between the RAT and the SW Amazonian Craton is likely, but rotated ~ 110° counterclockwise compared to its present position. On a broader scale, the RAT took part of the Umkondia megacontinent (Choudhary et al., 2019) during the late Mesoproterozoic, consistently positioned with the proposed plume center situated at the present northwestern Kalahari Craton as being the magmatic source for the presently WNW-ESE trend of the Rio Perdido dike swarm. After ca. 1100 Ma, the Umkondia rotated clockwise doking Laurentia and Baltica at ca. 1000–900 Ma.

A virtual geomagnetic pole obtained for the 1770–1760 Ma Caracol magmatic arc (Eastern Domain of the RAT) together with the 1780–1790 Ma Avanavero pole from the Amazonian Craton, allowed to

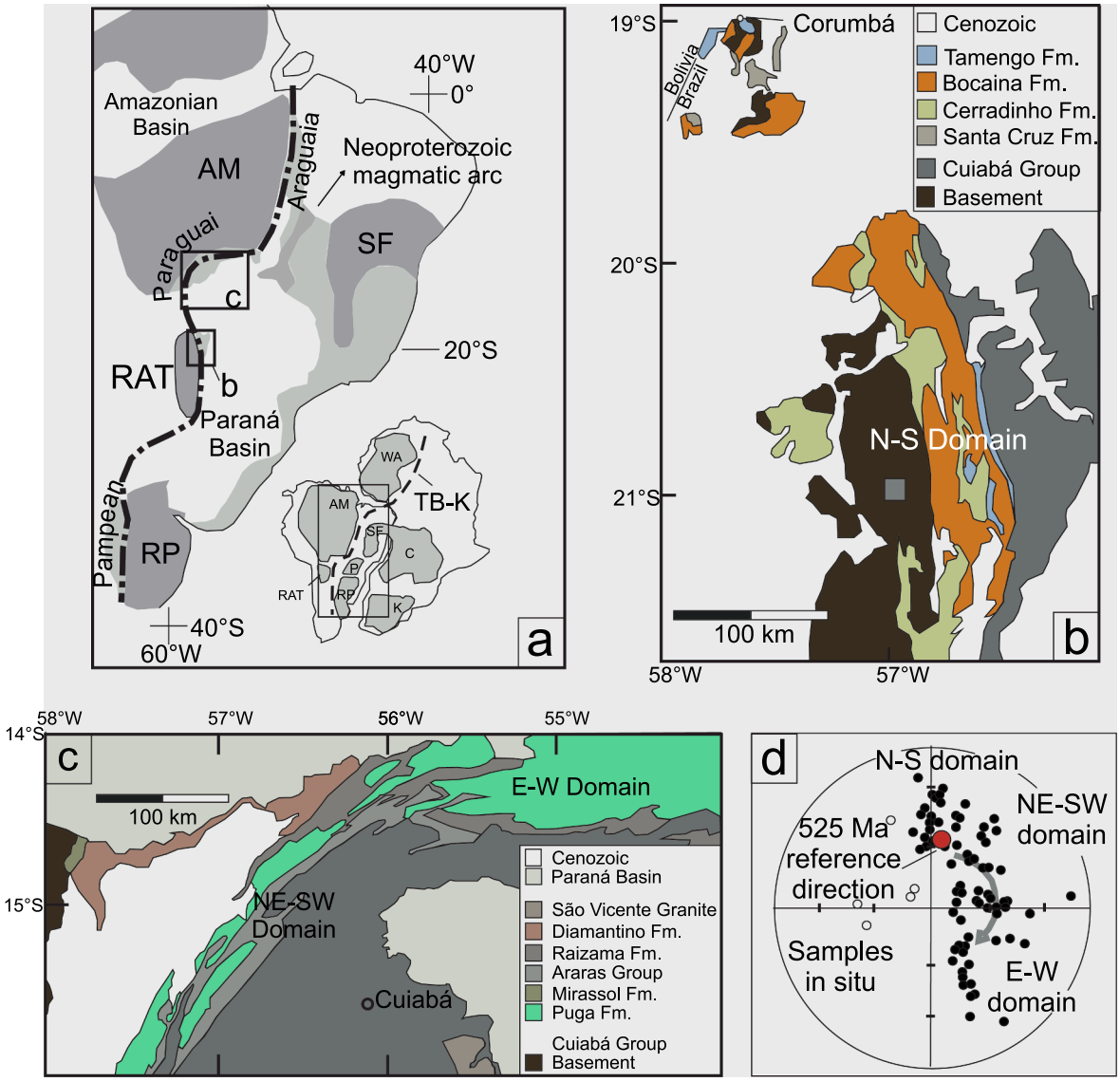


Fig. 14. a) South America showing the Neoproterozoic to Cambrian Paraguai-Araguaia and Pampean belts surrounding the Amazonian (AM), Rio Apa (RAT) and Rio de la Plata (RP) cratonic blocks. SF- São Francisco. Inset– Inferred suture zone (the Transbrasiliano-Kandi tectonic corridor (TB-K), Cordani et al., 2013) within the West Gondwana (C - Congo, WA - West Africa, K - Kalahari, P – Paranapanema). Boxes represent the geological maps of the N-S domain (b) and the NW-NE and E-W domains (c) of the Paraguai belt showing Ediacaran limestones and dolomites from the Bocaina Formation and Araras Group, respectively, which were paleomagnetically studied by Tohver et al. (2010). d) Distribution of the characteristic remanent directions from individual in situ samples from different strike domains forming a small circle arc indicated by the light gray arrow (after Tohver et al., 2010) (see text for details).

propose a possible link of the RAT–PT (in their present positions) and the Ventuari-Tapajós (VT) and Rio Negro-Juruena (RN) provinces at ca. 1780 Ma. Our reconstruction is consistent with the geologic, geochronologic and isotopic data from those provinces, which suggest the RAT and southwestern Amazonia (specifically, the PT), and possibly together with the VT and RN provinces evolved as a single tectonic unit during Paleoproterozoic (Teixeira et al., 2020, Ribeiro et al., 2020, 2022a, 2022b). At some time later, the RAT–PT block separated from proto-Amazonia and collided with it again at ca. 1340–1320 Ma along the Alto Guaporé Belt (e.g., Rizzotto and Hartmann, 2012; Ribeiro et al., 2022a, 2022b). During the formation of the Umkondia, Congo-São Francisco/Kalahari/India executed a transcurrent movement relative to Amazonia/West Africa (Choudhary et al., 2019). We suggest that it was characterized by a sinistral movement, and that after ca. 1380 Ma (see Choudhary et al., 2019), the RAT rotated counterclockwise relative to PT until their final collision with Amazonia at ca. 1320 Ma. Finally, the RAT likely rotated back during Gondwana assembly giving rise to the aulacogen Tucavaca Belt.

CRediT authorship contribution statement

Manoel S. D'Agrella-Filho: Conceptualization, Data curation, Formal analysis, Project administration, Writing – original draft, Writing – review & editing. **Ualisson D. Bellon:** Conceptualization, Formal analysis, Writing – review & editing. **Bruno V. Ribeiro:** Validation, Writing – review & editing. **Wilson Teixeira:** Validation, Writing – review & editing. **Ricardo I.F. Trindade:** Validation, Writing – review & editing. **Amarildo S. Ruiz:** Validating, Writing - review & editing and field work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.precamres.2024.107347.

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